

AKADEMIKA

WL-WSN

Wireless Sensor Network Laboratory

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. Digital Communication is one of the important subject need to teach while learning electronics.

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OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

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As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that its integrity is checked with respect to its checklist. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers or connecting chords are connected correctly as per experiment.

These kits must be employed only for the use for which it has been conceived, i.e. as educational equipment, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under following conditions,

- Ambient temperature: from 0 to 45 ° C.
- Relative humidity: from 20 to 80 %.

Avoid any immediate/significant change of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

INTRODUCTION

Wireless networks and wireless sensor networks have been a major research area in last two decades. Progress in wireless sensor Network in last couple of years is astonishing. There seems to be a lot of diversity in WSN applications tried. The real time field testing is very important for realistic and usable results. But to study wireless/sensor networks behaviors and performance by means of real time deployment or setting up a testbed is complex and requires much effort and financial resources. S-WIFI Net (Swift WIFI Network –testbed) design draws its inspiration from difficulties and bottlenecks experienced by researchers while using testbeds and simulators. S-WIFI Net is designed with objective of combining simulator environment features like configurability, comprehensive result collection, simple user interface, ease of operations and testbed features like hands on experience of real time wireless communication, data generation and field testing. User has been distanced from nitty-gritty of hardware programming and embedded system handling giving much ease and confidence in carrying out the experimentation and getting feel of real time behavior of wireless communication. S-WIFI Net design is aimed towards simple, user friendly and single station monitored testbed. Provision of in-built network layer operation transparent to user have reduced frequent hardware programming or reprogramming of nodes.

Swifinet Testbed

S-WiFiNet is a short-range wireless sensor network Testbed. The S-WiFiNet testbed is based on single module radio trans-receiver. The S-WiFiNet devices are battery operated and portable allowing the user to conduct indoor and outdoor experiment tests to study the wireless network behavior and communication patterns. Outdoor experiments allows student to study effect of environment and difficult terrain on the wireless communication. The details of network configuration parameters their significance and operation of the testbed is provided for the users in this manual. List of experiment and many other possibilities of experimentation is also discussed. The S- WiFiNet testbed is a very useful tool for understanding wireless devices, operation, and communication fundamentals. Wireless application developers can test the application before it is deployed on site.

S-Wifi Net Architecture

The S-WiFiNet testbed architecture is based on proprietary network stack S-WiFi. S-WiFi network stack is built on top of 802.15.4 MAC/PHY RFC for the 802.15.4 based trans-receiver microcontrollers. S-WiFi network stack is independent of the hardware platform (Processor). To demonstrate the hardware independence of the S-WiFi networkstack it is also ported on 802.11 based trans-receiver microcontroller. S-WiFi is based on fixed 16 bit addressing, two-tier hierarchical network. First layer device is MD (Master device). There is only MD in the network. Second layer devices are called as NDP (Network Processing Device). The NPDs acts as end node device as well router device. S-WiFi allows formation of up-to 5 hop, multi-hop network. The MD and NPD can attach up-to 30 devices in star network topology. S-WiFi network stack is self-organizing and self-healing network layer. It has two main components. A Master device which controls the network and device operations, emulating master sink node in real WSN application and End-Node cum Router device alternately referred as NPD (Network Processing Device). The NPD devices allow to communicate nodes beyond the range of master which can be cascaded to cover larger distances and more WSN nodes. The master is connected to laptop or desktop. A user-friendly GUI for configuration and operation of theSwifiNet testbed is provided. Each device is provided with LCD/OLED screen which displays the connectivity status, device configuration. The screen also displays additionalinformation as explained in later sections. SWiFi network layer stack starts self-organizing with master device is boot up. The device in the LQI Threshold vicinity of the master starts joining the master device. Once these devices are joined in the network in turn, they allow other devices in their vicinity to join in the network.

ESP32

WL-WSN Trainer is designed to study the WI-FI Wireless Technology IEEE 802.11 Standard and it covers all Topologies & Protocol. Trainer carrying a powerful WIFI Transceiver ESP-32-WROOM and it works with ESP32 plug-in. It transfers data using WI-FI Services from NPD and to Master. Trainer provides full accessibility to the user to Communicate between Master and Nodes. It is capable of Controlling WL-WSN Peripherals using GUI.

SPECIFICATION

General Specifications

1. Trainer should be a bench top model with an open PCB architecture.
2. Trainer should be implemented with a Wi-Fi module.
3. It includes on board peripherals which can be used for designing various small-scale applications.

Technical Specifications - System

1. The Trainer is based on Wi-Fi module, which is a powerful, generic Wi-Fi+BT+BLE MCU module Frequency range 2.4GHz ~ 2.5 GHz, Data rate: up to 150 Mbps.
2. System operates over battery.
3. **WSN Device with on-board Sensors**
 - OLED display 1.3"
 - LED indications: Power, Rx/Tx
 - Keys: Program, Reset, User key
 - Rechargeable battery power for 10 Hrs of operation
 - Serial interface for Programming
4. **Processor:**
 - 32-bit RISC CPU, up to 32MIPs
 - Data EEPROM with guaranteed 100k write operations
 - On Chip program memory 256 K
 - On Chip Data memory 32K RAM
 - 2 Nos of 10-bit ADC, 1 comparator
 - 5 x PWM (4 x timer, 1 x timer/counter)
 - 2 UARTs
 - SPI Master & Slave port with 3 selects
 - 2-wire serial interface
 - Watchdog timer
 - Radio 2.4 GHz IEEE 802.15.4 compatible.
 - Up to 30m range (PCB Antenna, Default)
 - Up to 500 m range (Ext antenna, On Order)
 - TX power 0 dBm, or +2.5dBm with external antenna
 - Receiver sensitivity -95dBm
 - TX current 15.3mA
 - RX current 17mA
 - 2.0-3.6V operation
5. **On Board Peripheral:**
 - **RELAY:**

It is an actuator, 5V Good SKY SPDT Mechanical Relay
Operate Time: 10 mSec. Max.

- **PWM:**
PWM signal to control the LED.
- **DAC:**
User can vary the potentiometer value and check the respective voltage at DAC connector.
- **ADC:**
Potentiometer: Potentiometer value 1K.
For generation of an analog signals.

- **BME Module:**

Temperature: -40 to 85°C.
Humidity: 0 to 100%.
Pressure: 300 to 1100 hPa.
Altitude: 0 to 30,000 ft.
I2C and SPI Interface.
3.3V operation.

- **Select DIO**

User can configure the external DIO on Sensor Interface Connector.

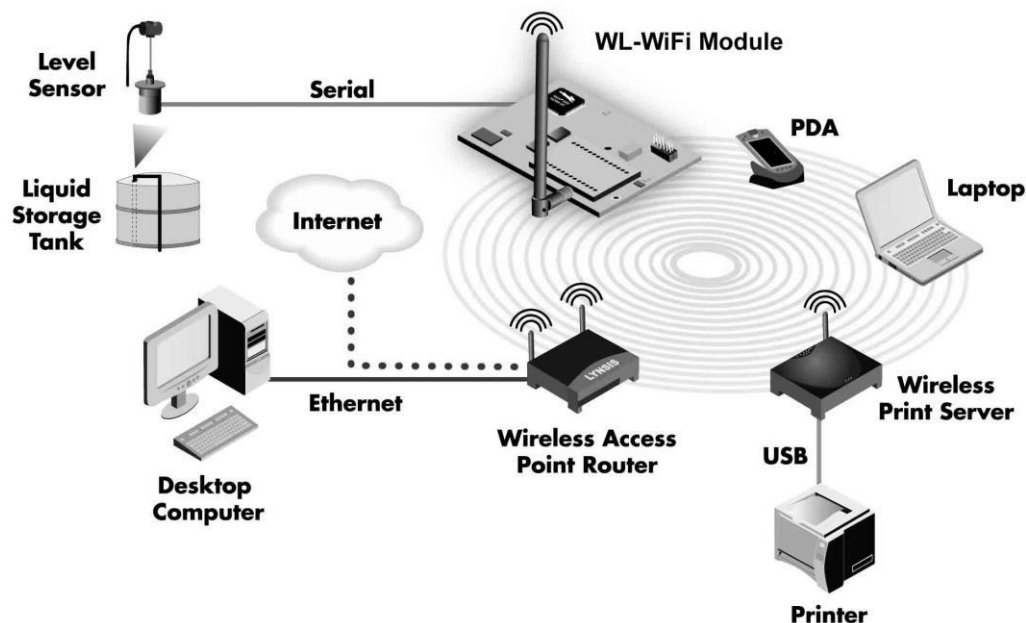
ACCESSORIES:

1	Master & Gateway (WAP) Module	01
2	Node's (NPD) Module	10
3	USB cable	06
4	Helix Antenna	12
5	Jumper wire Male to Female	10
6	Battery Charger	01

INTRODUCTION

Wi-Fi, also spelled Wi-Fi or Wii, is a [local area wireless technology](#) that allows an electronic device to exchange data or connect to the internet using 2.4 GHz [UHF](#) and 5 GHz [SHF](#) radio waves. A common misconception is that the term Wi-Fi is short for "*wireless fidelity*," however this is not the case. Wi-Fi is simply a trademarked phrase that means IEEE 802.11x.

Wi-Fi works with no physical wired connection between sender and receiver by using radio frequency ([RF](#)) technology, a frequency within the electromagnetic spectrum associated with radio wave propagation. When an RF current is supplied to an antenna, an electromagnetic field is created that then is able to propagate through space.



The cornerstone of any wireless network is an access point ([AP](#)). The primary job of an access point is to broadcast a wireless signal that computers can detect and "tune" into. In order to connect to an access point and join a wireless network, computers and devices must be equipped with wireless network adapters.

Wi-Fi is supported by many applications and [devices](#) including [video game consoles](#), home [networks](#), [PDAs](#), [mobile phones](#), major [operating systems](#), and other types of [consumer electronics](#). The [radio frequency](#) band used in Wi-Fi technology are 2.4GHz for [802.11b](#), [802.11g](#), or [802.11n](#), and 5GHz for [802.11a](#)

EXPERIMENT NO. 1

EXPERIMENT NO: 1

NAME: -

Introduction to Wireless Network Setup and Software.

OBJECTIVE: -

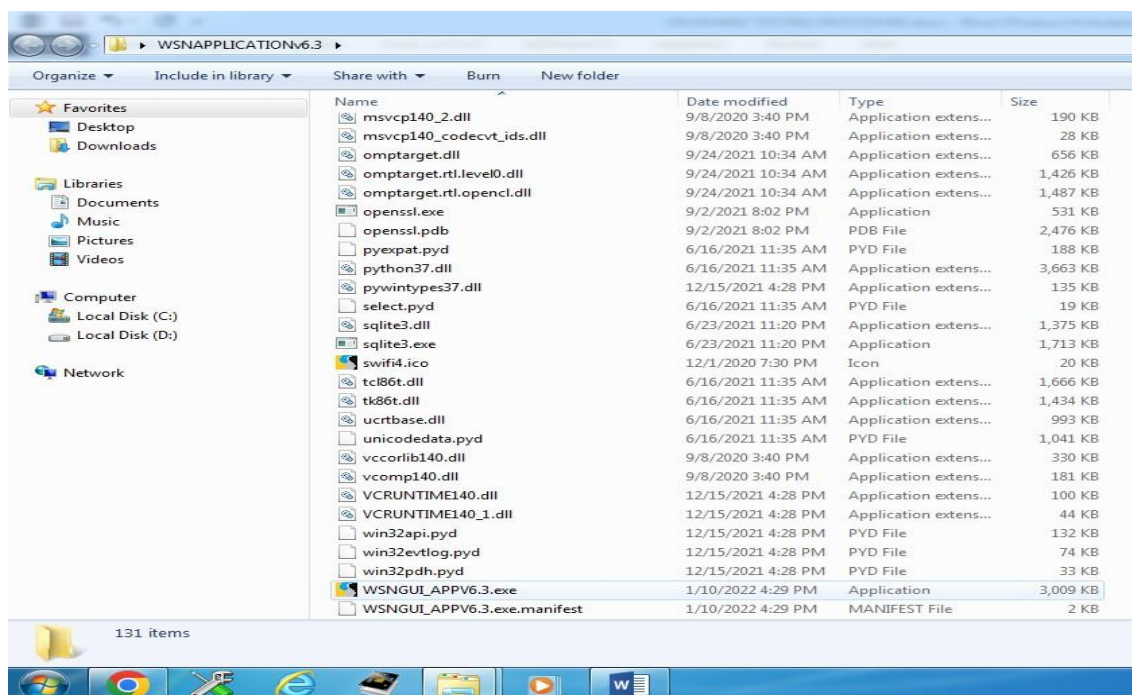
- To understand WSN main menu.
- To understand WSN Radio Parameter.
- To understand WSN Experiment Parameter.
- To understand WSN Experiment Control Command.

EQUIPMENTS: -

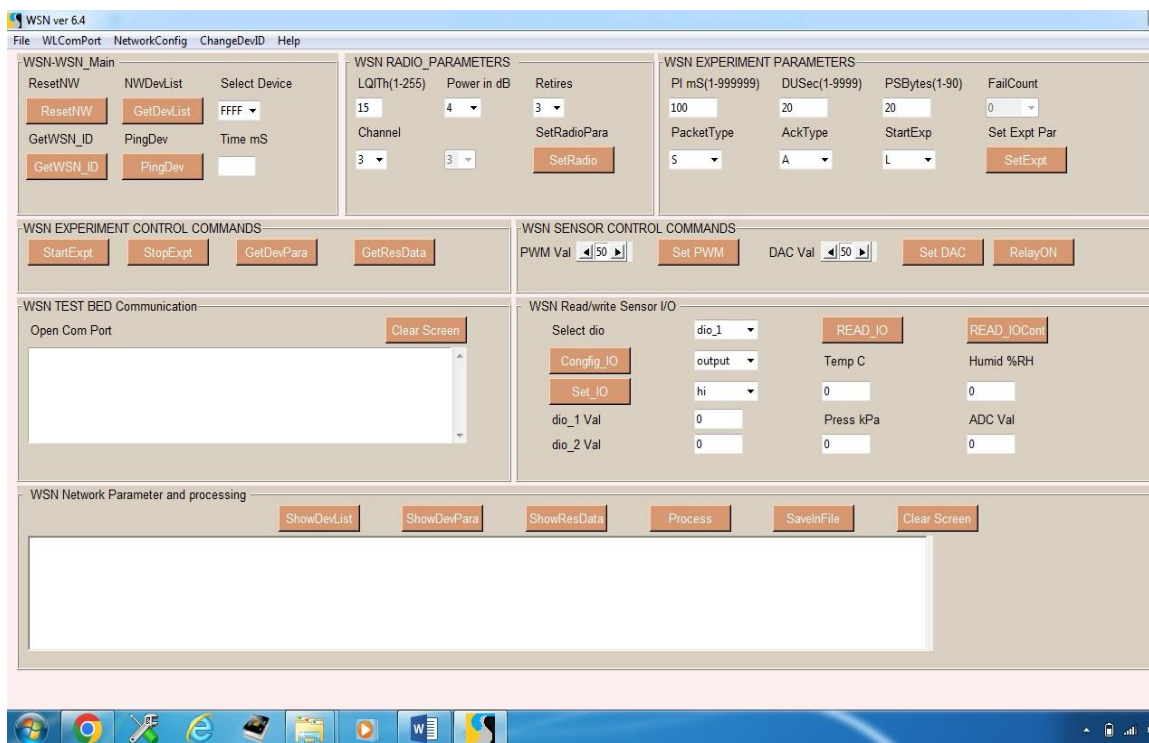
- WSN-IOT Master / Gateway Trainer 01 no
- WSN-IOT NPD 10 no's
- PC or Laptop with GUI Application software.

PROCEDURE: -

Open folder **WSNAPPLICATIONv6.3**.



Now open **WSNGUI_APPv6.3.exe** application file.



Main GUI commands: -



File

1. Exit

Exit the WSN GUI software User shall always exit the application using this command.

ComPort

1. Open Comport

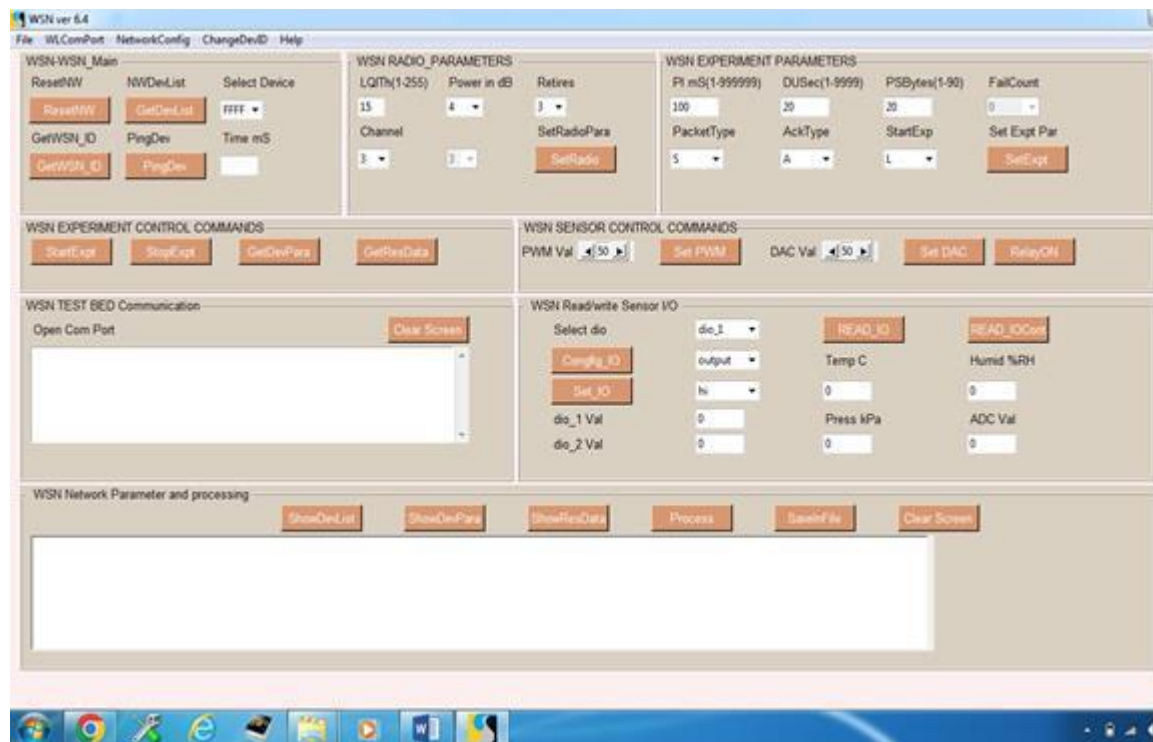
Open the serial communication port to which master device is connected.

2. Close Comport

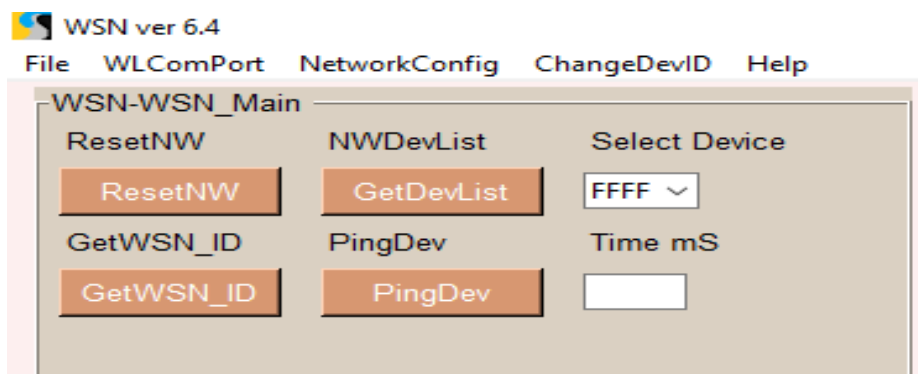
Close the open serial communication port.

Screens

1. WSN Main GUI screen



WSN-WSN MAIN (Network operation Commands)



a. Reset_NW

Pressing this will reset all the network devices. The Master and network devices will reset themselves. The network will enter into self-organizing mode. The devices will form network with Master as network **COORDINATOR**.

b. Get_DevList

This button queries master about the list of network devices connected in the network. The list is populated in **NW Devices** list Box. User can check the list

and select from the list for another device operation. Default list is 'FFFF' and '8000'. 'FFFF' indicates broadcast address. '8000' is master address.

c. Ping_Dev

User can select device form NW Devices list to ping the device. A packet with 50 bytes load is forwarded to the selected device. The device returns the packet to Master. The packet route time is shown in the Time box in milliseconds.

WSN RADIO PARAMETER.

WSN RADIO_PARAMETERS

LQIth(1-255)	Power in dB	Retires
15	4	3
Channel		SetRadioPara
3	3	SetRadio

- **Power in db.:** 0 to -32dB.
(Only available for high power modules).
- **LQIth (1-255):** 1-250 LQI Threshold.
Used by device to join the parent device. If the LQI seen By the device for the parent requesting join-me command is more than LQIth value then device accepts the join-me invitation.
- **Retires:** Retries 0 to 3 Transmissions.
If the AckType is set to 'H' or 'A' then device will try to re-send the packet if packet send failure occurs.

'SetRadioPara' : On Click these set parameters will be configured to the selected NPD.

WSN EXPERIMENTAL PARAMETER

WSN EXPERIMENTAL PARAMETERS

Pl mS(1-999999)	DUSec(1-9999)	PSBytes(1-90)	FailCount
100	20	20	0
PacketType	AckType	StartExp	Set Expt Par
S	A	L	SetExpt

- **PlmS (1-999999):** 1 to 999999 mS Packet Interval. Packet interval between data packets transmitted by the node.
- **DUSec (1-9999):** 1 to 9999 Seconds Experiment Duration. Duration for which node will generate data packets.
- **PSBytes (1-90):** 1 to 99 bytes Data size in Packet. Sets the number of data bytes in the data packet.
- **AckType:**
 - N= No ack.
 - H = Hardware ack.
 - A = Software ack from receiving device.
 - R = Node to Node Ack.

The device can transmit packet with this request to the destination device.
- **Packet Type:**
 - S = Sequential.
 - R = Random.

S: The packet generation at the node will be sequential with set Packet interval time

R: Packet Generation will be random within Packet interval time. Useful for testing random generated data packets.
- **Packet Transmission Start:** L = after start Command.
I = Immediate.
The data packets generation can be started with **Start** experiment command or they can be immediately started after the device is reset. Useful to study network congestion.

‘SetExptPara’ these parameters will be configured / set to the selected device from the NW device list by pressing this button.

WSN EXPERIMENTAL CONTROL COMMAND.**a. StartExpt**

User can start the Data packets generation from each of the network devices. The experimental parameters are as explained above. This is effective if the Packet Transmission Type is configured as 'L'

Selecting 'FFFF' will select broadcast address so that all the devices can generate data packets. Selecting specific device address will allow only selected device to start generating Data Packets. Selecting '8000' will have no effect as Master cannot generate the data packets.

b. StopExpt

User can stop the Data packets generation from each of the network devices before the experiment duration is over. This is also aborting the data packet generation. Selecting 'FFFF' will select broadcast address so that all the devices can stop generate data packets. Selecting specific device address will allow only selected device to stop generating Data Packets. Selecting '8000' will have no effect as Master cannot generate the data packets.

c. GetAllPara

User can read the radio and experimental configuration from the selected device from the NW device list. In this case user cannot select 'FFFF' address is selected it will select '8000' address by default. The experiment parameters are not applicable to Master device. The data collected using this command will be used for generating tabulated data.

d. GetExpData

User can read Total Transmitted/Received packets, Transmitted/Received control packets, Total data packets, failed data packets, LQI (Avg, Max, Min) Transmission time (Avg, Max, Min) etc. from each of the device from the selected device from the NW device list. In this case user cannot select 'FFFF' address is selected command will be sent to '8000' address by default. The data collected using this command will be used for generating tabulated data.

WSN Network Parameter and Processing

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
5001	01	01	129	8000	00	129	----	--	---
5005	01	02	069	5001	01	000	----	--	---
5003	01	02	090	5001	01	000	----	--	---
5002	01	02	084	5001	01	000	----	--	---
5004	02	03	108	5002	02	108	5005	02	165

ShowDevList

Pressing this button will show the network connectivity map in the Experiment Process Result Parameter window. User should have collected the device list from the master device by pressing 'GetDevList' button.

The device will always try to join a parent device meeting LQIth value and minimum hop from the master.

Dev: Device Address of the device joined in the network.

- **Parents:** Number of parents' device received joined requests. Device can have at the most two parents.
- **Hops:** Number of hops the device is away from the Master device. It is hop+1 away from primary (first) parent.
- **DevLQI:** LQI value seen by the device towards first parent.
- **Parent1:** Address of the primary parent to which device has joined. In normal case all the communication towards master will pass through this device.
- **P1Hop:** Parent1 hops from the master.
- **P1LQI:** LQI value seen by the parent1 towards joined device.
- **Parent2:** Address of the secondary parent to which device has joined. In case primary device does not respond the device use this address as alternate path towards master device.
- **P2Hop:** Parent1 hops from the master.
- **P2LQI:** LQI value seen by the parent1 towards joined device.

ShowDevPara

Pressing this button will show the configured/set radio and experiment parameters in the Experiment Process Result Parameter window. User should have collected these parameters by pressing 'GetAllPara' button for each device in the NW device list.

WSN NW WSN_Analysis

ShowNetMap ShowInfo ShowResData Process

WSN Network Parameter and processing

Experiment Process Result Parameters

Clear Screen

Dev	PktIntvl	Dur	ExpDPkt	PktSize	Retry	AckType	Seq/Rand	StartExp	LQIth	Power	CA_val	BE_val	FC
8000	000100	0010	0	80	03	H	S	L	090	00	03	03	00
5001	000010	0005	500	80	01	H	S	L	050	03	03	03	00
5005	000010	0005	500	80	01	H	S	L	050	03	03	03	00
5003	000010	0005	500	80	01	H	S	L	050	03	03	03	00
5002	000010	0005	500	80	01	H	S	L	050	03	03	03	00
5004	000010	0005	500	80	01	H	S	L	050	03	03	03	00

Dev: device address of the device joined in the network

- **PktIntvl:** Packet interval as explained in radio and experiment parameter section in this document.
- **Dur:** Experiment duration in seconds.
- **ExpDPackets:** Expected data packets generated by the node device.
It is Duration in seconds/ PI in mS.
- **PktSize:** As explained in radio and experiment parameter section.
- **Retry:** As explained in radio and experiment parameter section.
- **AckType:** As explained in radio and experiment parameter section.
- **Seq/Ran:** As explained in radio and experiment parameter section.
- **StartExp:** As explained in radio and experiment parameter section.
- **LQIth:** As explained in radio and experiment parameter section.
- **Power:** As explained in radio and experiment parameter section.

ShowResData

Pressing this button will show the parameters obtained from each device in the Experiment Process Result Parameter window. User should have collected these parameters by pressing 'GetExpData' button for each device in the NW device list.

WSN NW WSN_Analysis

ShowNetMap ShowInfo ShowResData Process

WSN Network Parameter and processing

Experiment Process Result Parameters

Clear Screen

Dev	TxPkt	RxPkt	RxTxDPkt	TxFPkt	TxFDPkt	TxCNPkt	RxCNPkt	NwStTm	TxAAtm	TxMxTm	TxMnTm	LQIA	LQIMx	LQIMn
8000	0006	1063	1052	0001	0000	0006	0011	0000	0000	0000	4095	123	135	054
5001	1075	0626	0490	0017	0002	0583	0022	0254	0044	0287	0003	084	132	027
5005	0299	0010	0122	0174	0172	0005	0010	0293	0128	0415	0007	108	162	039
5003	0326	0012	0175	0147	0146	0005	0012	0858	0141	0521	0078	102	135	075
5002	0396	0193	0294	0097	0092	0010	0012	0922	0005	0521	0004	102	114	075
5004	0506	0010	0498	0002	0002	0006	0010	2743	0011	0072	0003	087	168	024

Dev: device address of the device joined in the network.

- **TxPkt:** Total transmitted packets.
- **RxPkt:** Total received packets.
- **RxTxDPkt:** Total received or transmitted Data packets.
- **TxFPkt:** Failed transmitted packets.
- **TxFDPkt:** Failed transmitted data packets.
- **TxCNPkt:** Total transmitted control packets.
- **RxCNPkt:** Received Control packets.
- **NWStTm:** Network set up time for the device (Time taken by the Device to join the network).
- **TxATm:** Average Packet transmission time in mS.
- **TxMnTm:** Minimum Packet transmission time in mS.
- **TxMxTm:** Maximum Packet transmission time in mS.
- **LQIA:** Average LQI of the transmitting device seen by this device.
- **LQIMx:** Maximum LQI of the transmitting device seen by this device.
- **LQIMn:** Minimum LQI of the transmitting device seen by this device.

Process

Pressing this button will show analysis of the data obtained in previous windows in the Experiment Process Result Parameter window. User should have collected these parameters by pressing 'GetAllPara' and 'GetExpData' button for each device in the NW device list.

The screenshot shows the 'WSN Analysis' window with the 'Process' button highlighted. Below it, the 'Experiment Process Result Parameters' table is displayed. The table has 8 columns: Dev, RxTxDPkt, TDPkt, CommLoad%, Dev Effi%, NwExpEffi%, and NwEffi%. The data is as follows:

Dev	RxTxDPkt	TDPkt	CommLoad%	Dev Effi%	NwExpEffi%	NwEffi%
8000 1052	10	10	1.616	10	42.08	66.624
5001 0490	500	500	123.469	98.0	10	10
5005 0122	500	500	12.295	24.4	10	10
5003 0175	500	500	9.714	35.0	10	10
5002 0294	500	500	17.483	58.8	10	10
5004 0498	500	500	3.213	99.6	10	10

Dev: device address of the device joined in the network.

- **RxTxDPkt:** Total received or transmitted packets.
- **TDPkt:** Expected Data packets to be transmitted. It is Duration in seconds/ PI in mS.
- **CommLoad%:** It is $(RxCNPkt + TxCNPkt) / RxTxDPkt$.
- **Dev Effi%:** It is $(ExpDPackets \text{ by the device}) / RxTxDPkt$.
- **NwExpEffi%:** It is $(\text{sum of ExpDPkts by all the devices}) / (RxTxDPkts \text{ by the Master})$.
- **NwEffi%:** It is $(\text{sum of RxTxDPkts by all the devices}) / (RxTxDPkts \text{ by the Master})$.

Master).

Note: This can be more than 100% as the master can receive more packets due to retransmission of the data packets due to failure at transmitting device side.

NetworkConfig > NetConfig

NetworkConfigSetting

SetGet FLT DevList

SrcDev: FFFF NoDev: 0

Dev1: Dev2: Dev3: Dev4: Dev5: Dev6: Dev7: Dev8: Dev9: Dev10:

GetFltList SetList

GetFltList SetList

Control FLT DevList

Apply: N Allow/Block: A

FltLstControl Exit

SetFltControl Exit

SrcDev – Enter the device address for which you want to set as a source device.

NoDev – Enter the number of device you configured (Eg. Dev1 =1009 Dev2 = 1003 So the NoDev =2)

Dev1- Dev10 – Enter the Device address of the Nodes which you want to connect to SrcDev .

GetFltList- Get the list of device connected to the SrcDev . Simply put the device address of which you wish to see in SrcDev and click on SetFltControl to get the devices connected .

SetList – As you have stated the desired network configuration

SetFltControl- Confirms the given network formation.

Apply – N/Y (Yes/No)**A-** Allow.**B-** Block.**WSN SENSOR CONTROL COMMAND**

The Sensor GUI screen can be selected as discussed in Main GUI command set.

Read_IO

Reads sensor data from the selected device from NW Devices list. This can be one time read. The data will be populated in WSN Read Sensing I/O window.

Read_IOCont-

Reads sensor data from the selected device from NW Devices list. This will be taking Continuous reading . The data will be populated in WSN Read Sensing I/O window.

Temperature

Shows data in raw format from ambient temperature sensor.

Humidity

Shows data in raw format from ambient Humidity sensor.

Pressure

Shows data in raw format from ambient Pressure sensor.

LEDPWM Value

User can select PWM setting from 0 to 99 from this box.

Set PWM

The set value in the LEDPWM value will be sent to the selected device from the NW devices list.

RELAY

User can switch ON / OFF the relay.

DAC

User can vary the potentiometer value and check the respective voltage at DAC connector.

Select dio

User can configure the external DIO on Sensor Interface Connector.

Config_IO-

User must configure the External / additional Sensors by selecting the dio and whether connected to Sensor interface.

Set_IO –

Sets high or low for the given selected sensor (dio_1 or dio_2).

dio_1 val or dio_2 val –

Readings from respective external Sensors.

EXPERIMENT NO. 2

EXPERIMENT NO: 2

NAME: -

Measurement of Packet Delivery Ratio & Communication Overload.

EQUIPMENTS: -

- WL-WSN NPD 6-10 no.
- Master / Gateway Trainer Kit 01 no
- PC or Laptop with WSN Application software.

THEORY: -

Packet Delivery Ratio

Packet delivery ratio is the ratio of number of packets received at the destination to the number of packets sent from the source.

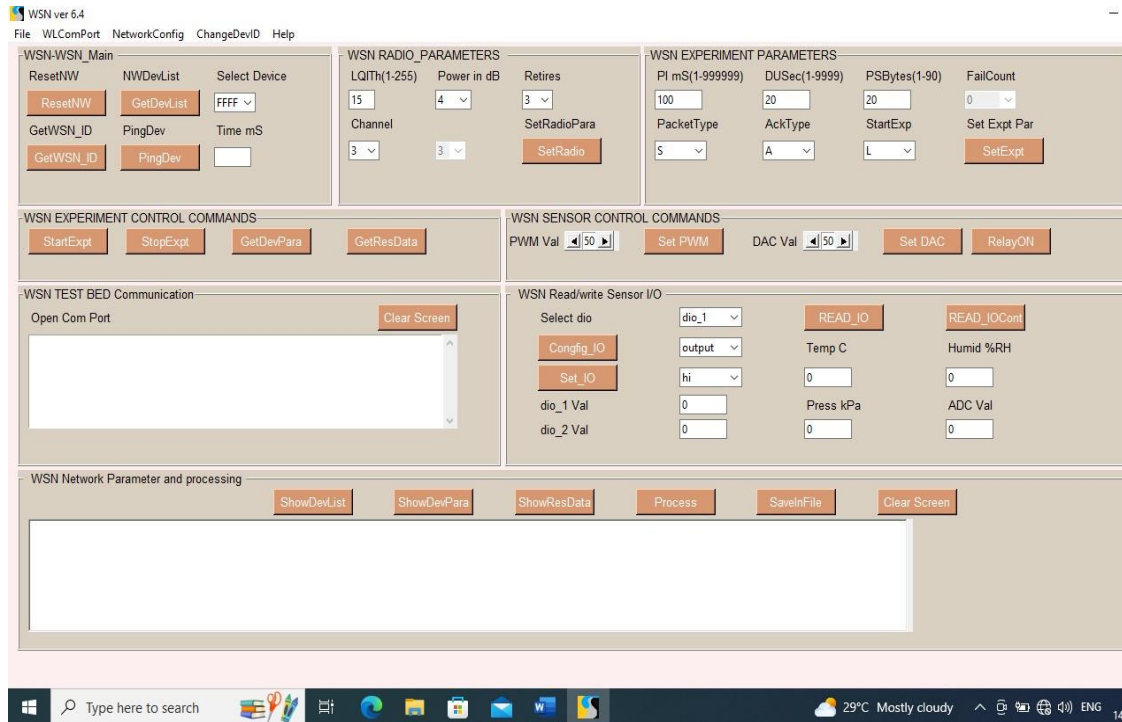
Packet delivery ratio is a very important factor to measure the performance of routing protocol in any network. The performance of the protocol depends on various parameters chosen for the simulation. The major parameters are packet size, no of nodes, transmission range and the structure of the network. The packet delivery ratio can be obtained from the total number of data packets arrived at destinations divided by the total data packets sent from sources. In other words, Packet delivery ratio is the ratio of number of packets received at the destination to the number of packets sent from the source. The performance is better when packet delivery ratio is high.

Communication load

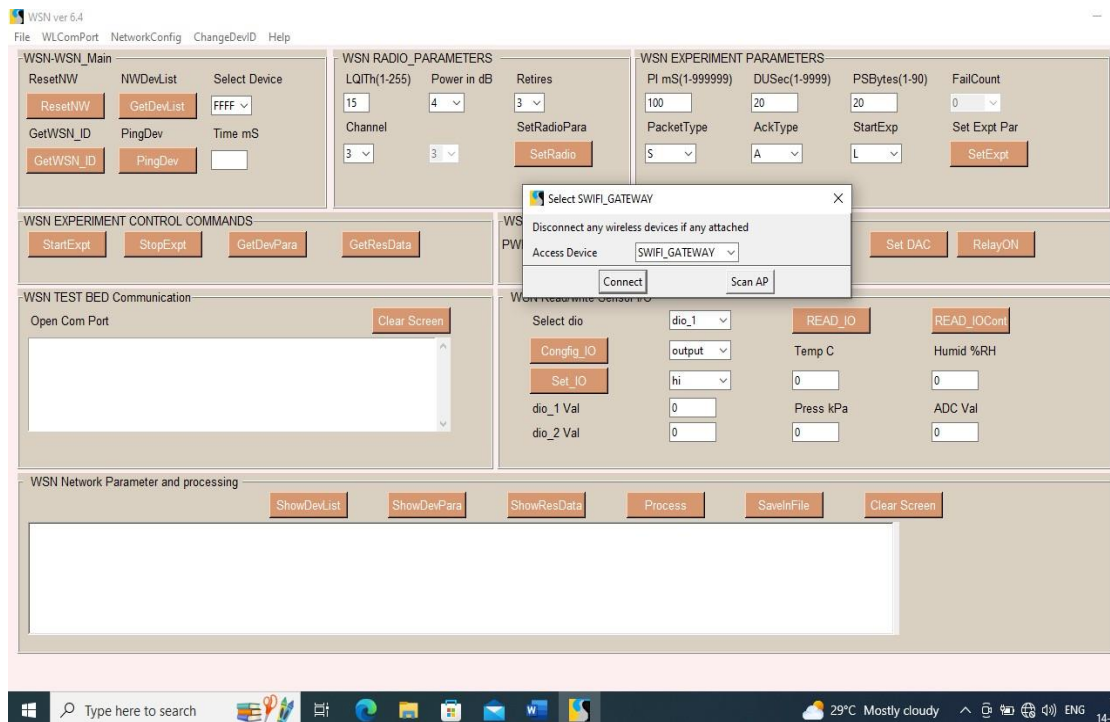
Communication load refers to the amount of data transmitted over a network within a specific time period. It is a measure of the burden imposed on the network by the data being sent or received. Communication load can be expressed in terms of data rate (amount of data transmitted per unit of time) or total traffic volume (the overall amount of data transmitted). Managing communication load is crucial for optimizing network performance, particularly in scenarios where bandwidth is limited or energy efficiency is a concern.

PROCEDURE: -

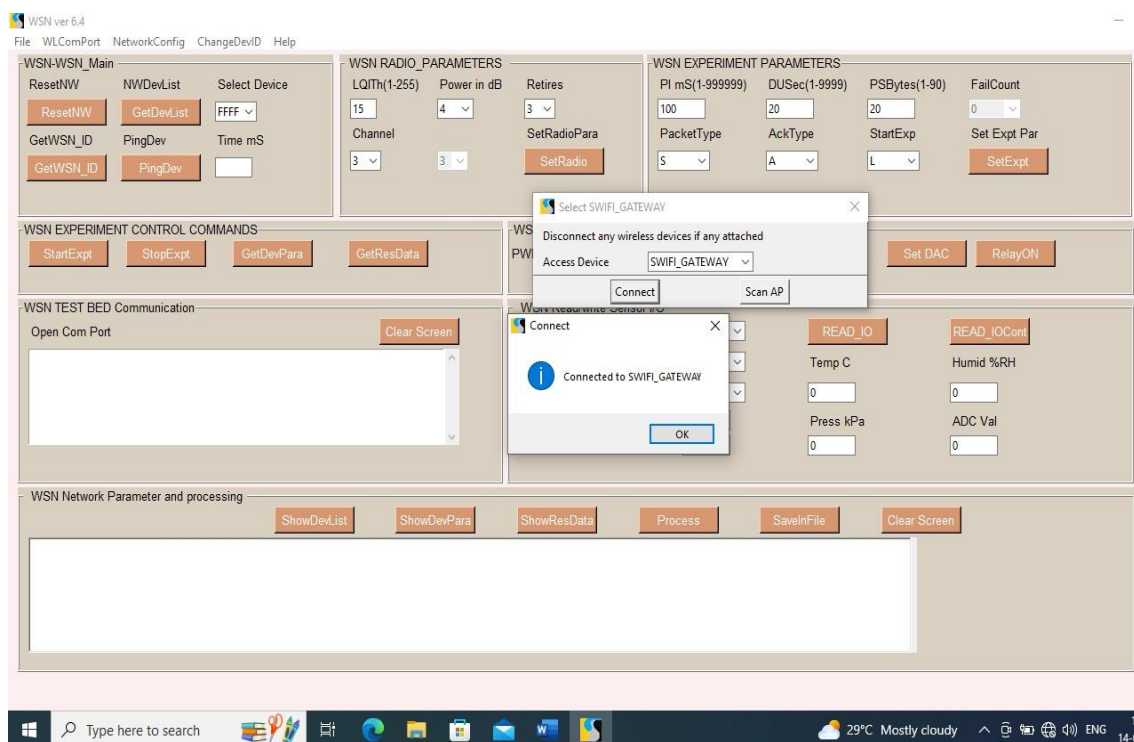
- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application.exe



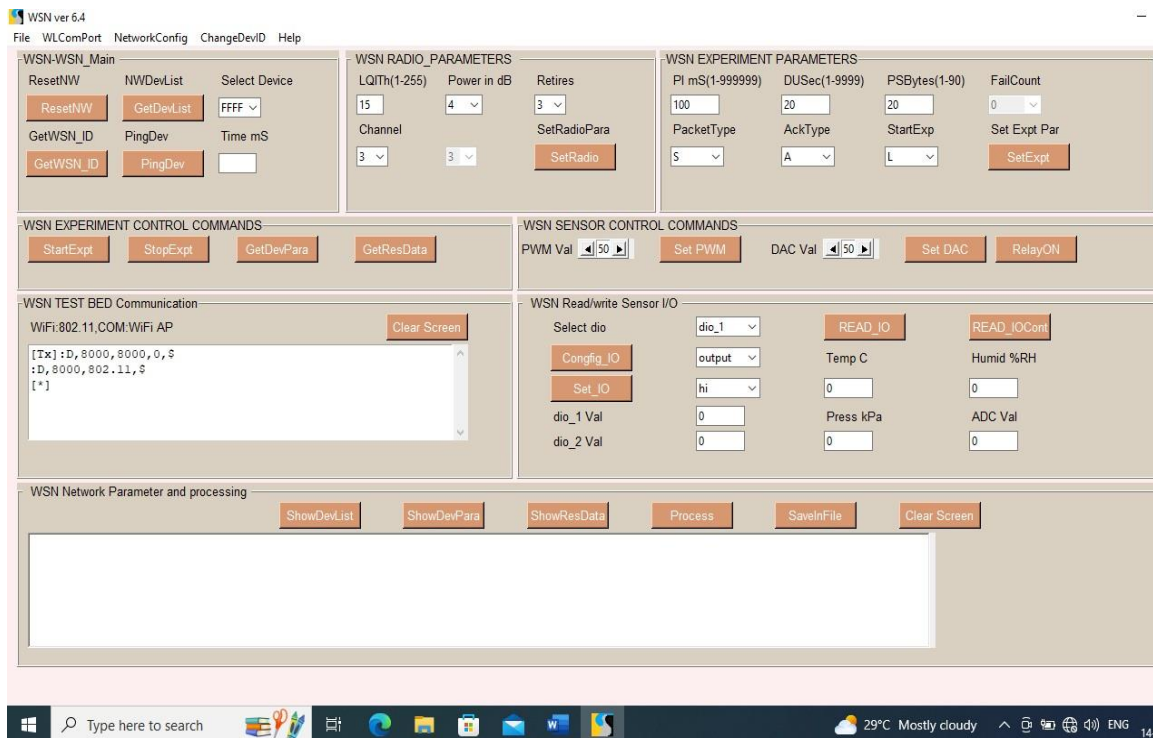
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.



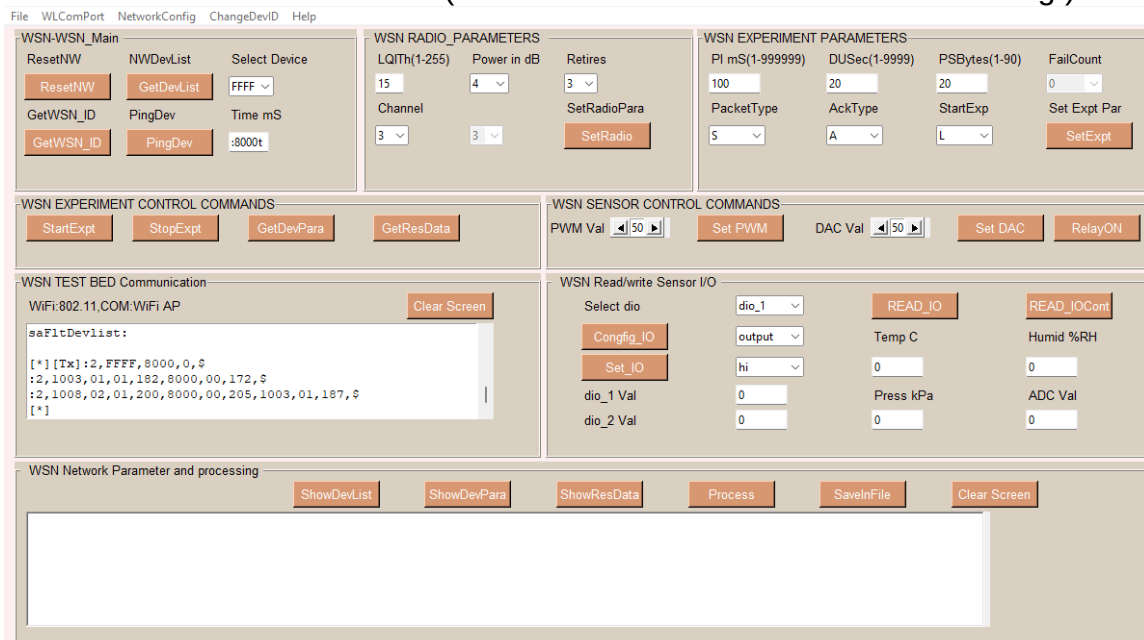
- Click on **Connect** tab.



- Click **OK**.
- Now We get the COM Port and Master device address on **WSN TEST BED Communication**.



- Now switch on 2 of the nodes .
- Before starting any experiment , always Reset the Network from the Master Device to clear all previous data.
- On the OLED Display of master device you will see the No. of Nodes Connected .
- Now click on **Get Device List** (WSN-WSN_Main) and wait for a while and then click **Show Device List** (WSN Wireless Parameter and Processing) .



File WLCmPort NetworkConfig ChangeDevID Help

WSN-WSN_Main
ResetNW NWDDevList Select Device
ResetNW GetDevList FFFF
GetWSN_ID PingDev Time mS
GetWSN_ID PingDev :8000t

WSN RADIO_PARAMETERS
LQIth(1-255) Power in dB Retires
15 4 3
Channel SetRadioPara
3 3 SetRadio

WSN EXPERIMENT PARAMETERS
PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
100 20 20 0
PacketType AckType StartExp Set Expt Par
S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS
StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS
PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication
WiFi:802.11.COM:WiFi AP Clear Screen
saFltDevList:
[*] [Tx]:2,FFFF,8000,0,\$
:2,1003,01,01,182,8000,00,172,\$
:2,1008,02,01,200,8000,00,205,1003,01,187,\$
[*]

WSN Read/write Sensor I/O
Select dio dio_1 READ_IO READ_IOCont
Config_IO output Temp C Humid %RH
Set_IO hi 0 0
dio_1 Val 0 Press kPa ADC Val
dio_2 Val 0 0 0

WSN Network Parameter and processing
Network Device List ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

Dev	Parents	Hops	Dev LQI	Parent1	P1 Hop	P1 LQI	Parent2	P2 Hop	P2 LQI
1003 01	01	182	8000	00	172	----	----	----	----
1008 02	01	200	8000	00	205	1003	01	187	

- Now we will get the list of nodes connected to the network.

Dev	Parents	Hops	Dev LQI	Parent1	P1 Hop	P1 LQI	Parent2	P2 Hop	P2 LQI
1003	1	1	182	8000	0	172	----	--	---
1008	2	1	200	8000	0	205	1003	01	187

Set these experimental Parameter for Experimentation.

For Number of Nodes_02

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L

- Now click on SetExpt.
- Then click on the GetDevPara on the (WSN EXPERIMENT CONTROL COMMAND)
- Now Click on the ShowDevPara (WSN Wireless Parameter and Processing). Here we can now see the display of all the device parameters that we have set up for experimentation.
- Then click on the StartExpt. (WSN EXPERIMENT CONTROL COMMAND)
- Now the packet communication from the nodes will start and end after the specified time. (in this case 20 sec.)
- Check and wait for the maximum Duration (DU sec) value plus 10 sec then click on GetResData. (WSN EXPERIMENT CONTROL COMMAND)

- Then click on the ShowResData. (WSN Wireless Parameter and Processing). Here we can now see the display as shown:

Dev	Tx Pkt	Rx Pkt	RxTxDPkt	TxF Pkt	TxFDPkt	TxCN Pkt	RxCN Pkt	NwStTm	TxA Tm	TxMxTm	TxMnTm	LQIA	LQIMx	LQIMn
8000	1221	3408	2108	7	0	1221	1300	0	0	0	4095	121	157	90
1003	611	63	588	0	0	23	63	2344	4	534	3	142	155	132
1008	2577	236	2411	18	0	166	236	2327	6	651	2	141	160	110

- Then click on the Process for final data. All these can be saved on a Notepad as txt files and then import it to excel for further processing and plotting

Result Table:

Dev	RxTxDPkt	TDPkt	CommLoad%	Dev Effi%	NwExpEffi%	NwEffi%
8000	3184	0	53.989	0	796	26
1003	1228	40	19.3	3070	0	0
1008	1226	40	20.881	3065	0	0

Now reset the network and perform same experiment with 4 number of nodes

Number of Nodes- 4

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L

Now reset the network and perform same experiment with 6 number of nodes

Number of Nodes - 6

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L

Packet Delivery Ratio:

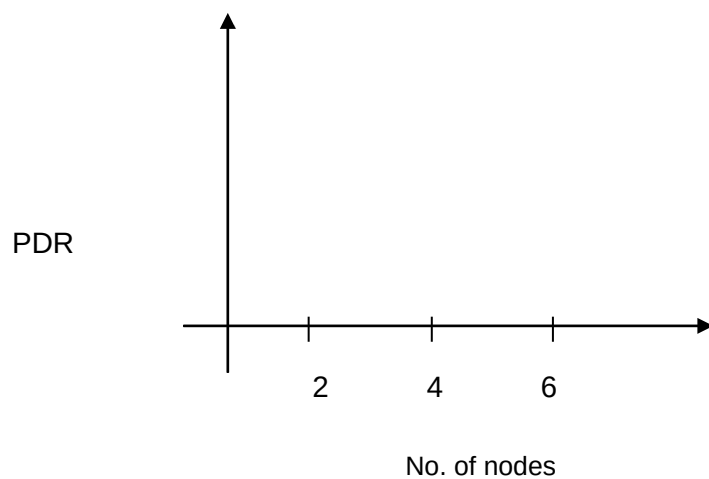
Formula:

Packet Delivery Ratio=Number of Packet Received by master / Number of Packet send by node

$$\text{PDR} = \text{RxTxDPkt (Master)} / (\text{Sum of all nodes RxTxDPkt})$$

Now calculate average packet delivery ratio for each set of experiment i.e. Nodes 2,4,6.

Now Plot the Graph of PDR v/s No of Nodes.

**Communication Load**

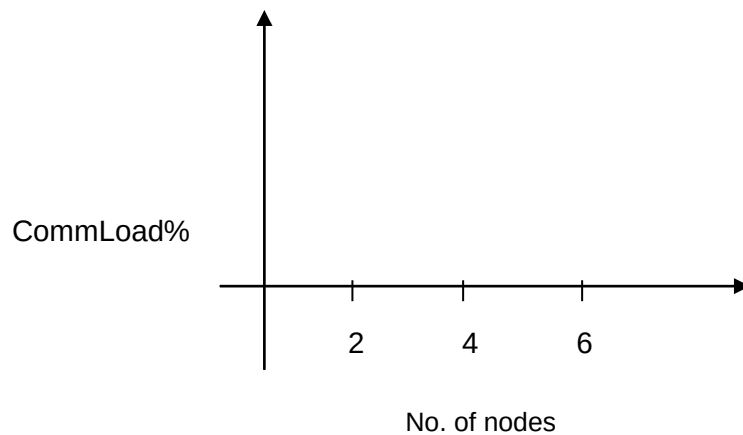
Formula:

CommLoad%: It is $(\text{RxCNPkt (Master)} + \text{TxCNPkt (Master)}) / \text{RxTxDPkt (Master)}$

CommLoad% for each node is calculated by the software and is available in the Process (Result Window).

Now calculate average CommLoad% for each set of experiment i.e Nodes 2,4,6.

Now Plot the Graph of CommLoad% v/s No of Nodes.



Note: You can do the same set of experiment by varying ACKNOWLEDGEMENT Type to : N (no ack) & S(Software ack) and keeping other parameters constant.

After every experiment reset the network for proper reading.

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L
2.	100 ms	20	20	S	A	L

Conclusion-

- a) As we increase the No. of nodes we observe a drop in PDR as the network becomes overloaded.
As we increase the No. of nodes the communication load

EXPERIMENT NO. 3

EXPERIMENT NO: 3

NAME: -

Measurement of Distance and Signal Strength (LQI) & find relation between them.

OBJECTIVE: -

- Measure the Distance and Signal Strength (LQI).
- Find the Relation between them.

EQUIPMENTS: -

- WL-WSN NPD's 01 no's
- Master / Gateway Trainer Kit 01 no
- PC or Laptop with WSN Application software.

THEORY: -

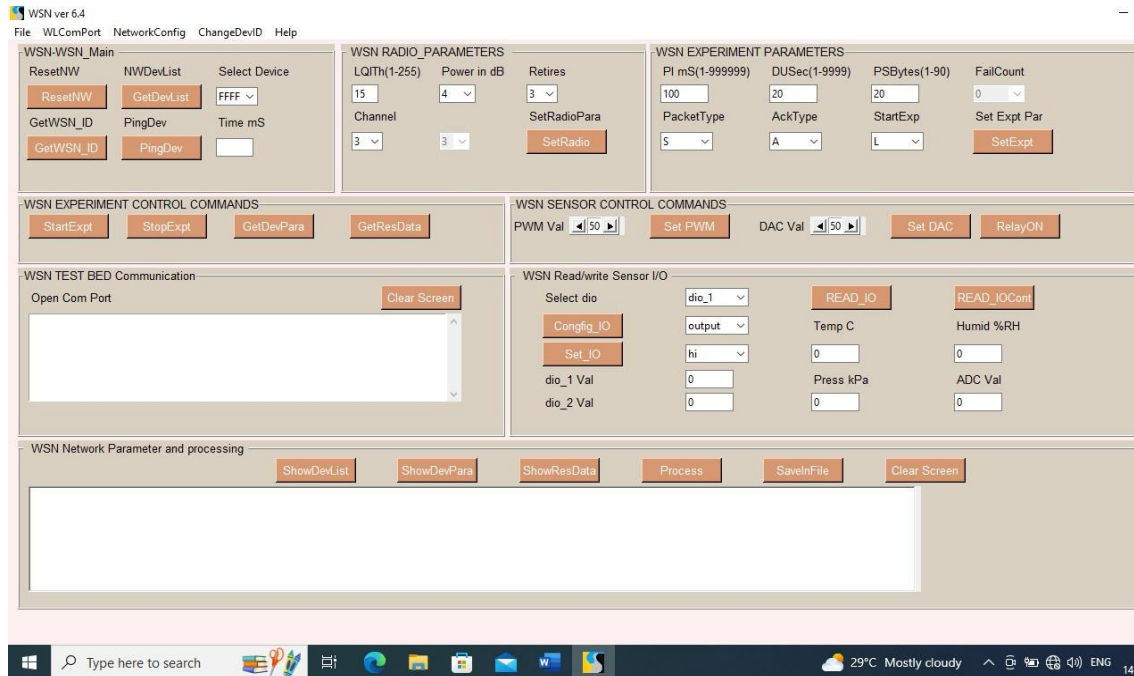
An LQI-Based Ranging Technique .

Ranging technology which estimates the distance between two communicating wireless nodes has been widely used as a necessary component in localization solutions for wireless sensor networks (WSN). LQI (Link Quality Indicator) is a metric introduced in IEEE 802.15.4 that measures the error in the incoming modulation of successfully received packets that pass the CRC (Cyclic Redundancy Check). Because of low system cost and less computational complexity, LQI-based ranging techniques are increasingly applied in Zigbee sensor networks. However, due to the environmental affects and electronic noise generated by hardware, raw LQI data could not be directly aligned with distances. To eliminate errors in LQI data and obtain higher ranging accuracy, we design and evaluate a novel LQI-based ranging technique which includes three essential data processing components: pre-correction, error compensation and mixed regression analysis. First, anchor nodes with known locations are used in pre-correction process to correct LQI measurements against the empirical regression function obtained from historical data. Then, error compensation is applied to eliminate the intrinsic error in LQI data. Finally, ranging results are refined by the mixed regression analysis. The proposed ranging technique is implemented and evaluated on a Zigbee sensor prototype– Tarax. Experiment results show that the average ranging error is less than 1m, confirming that the proposed technique is able to achieve higher ranging accuracy and suitable for localization applications in WSN.

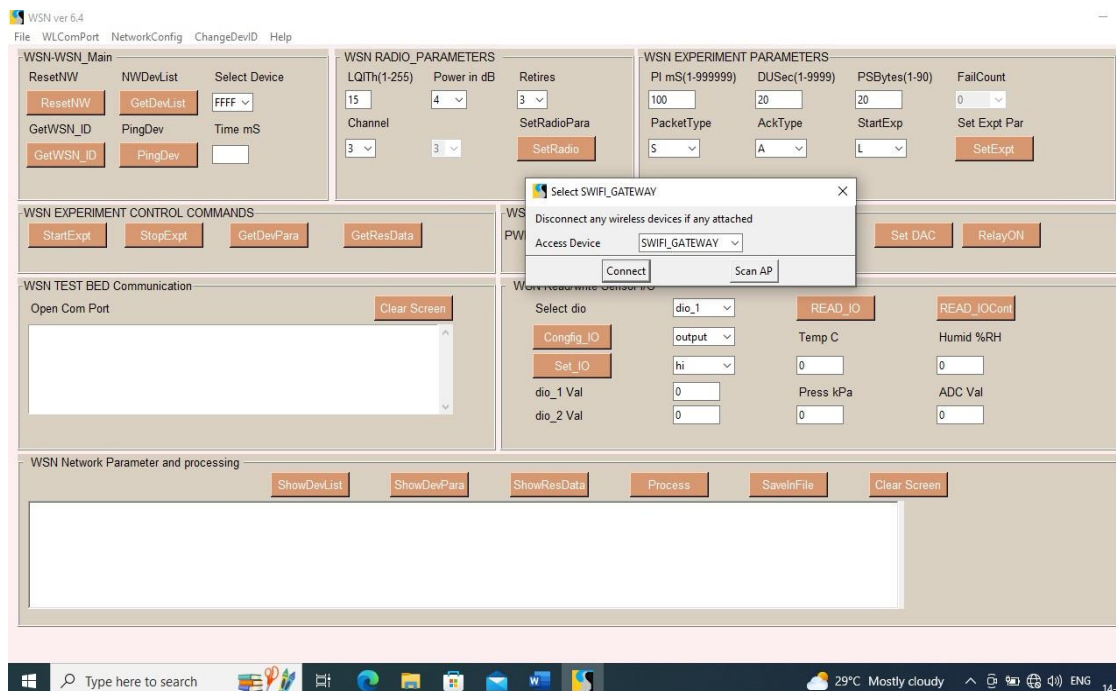
Keywords: Link quality indicator (LQI); ranging techniques; Zigbee; wireless sensor networks; distance estimation.

PROCEDURE: -

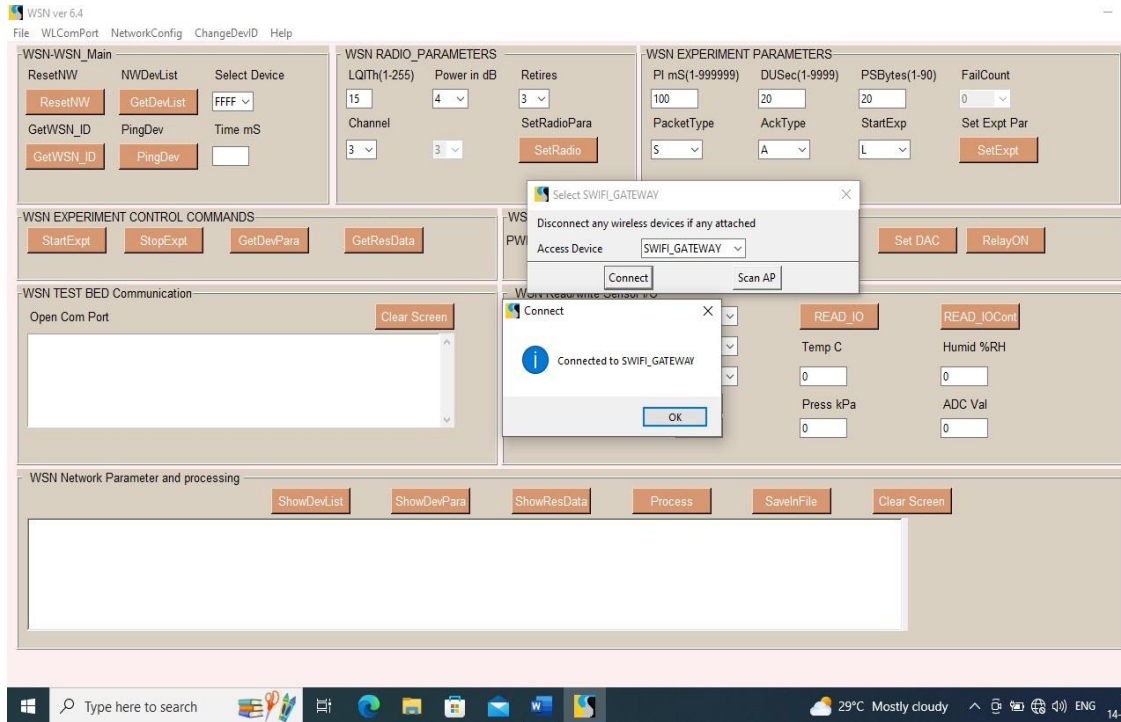
- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.



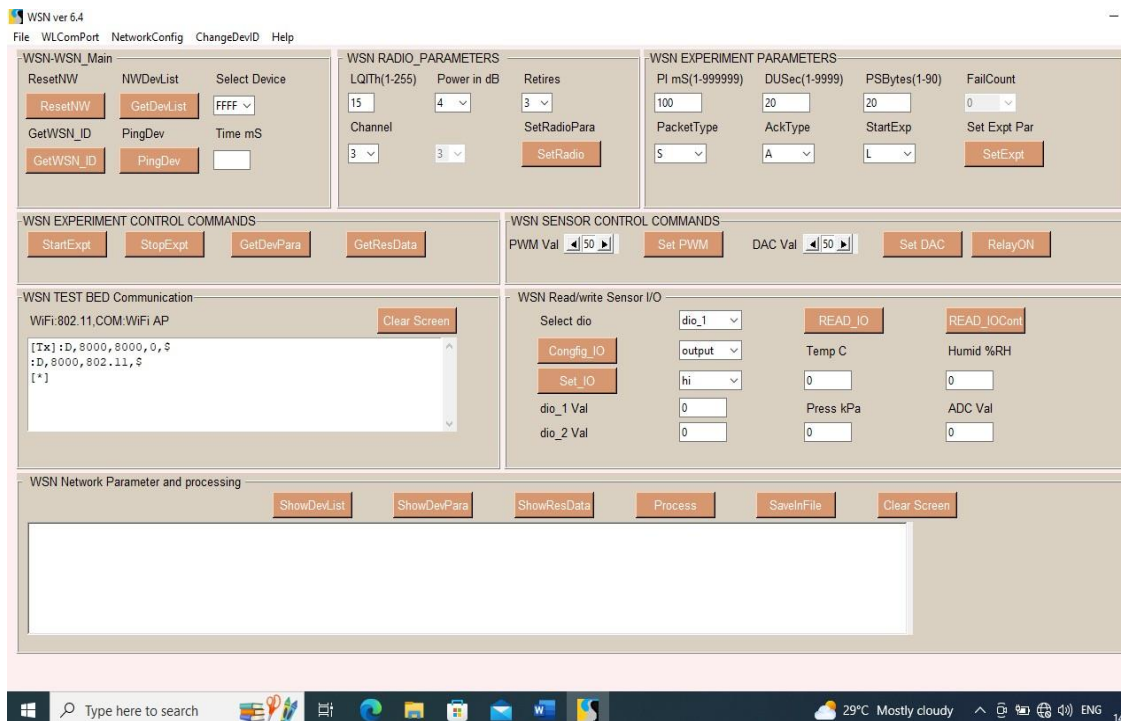
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.



- Click on **Connect** tab.



- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.



- Now switch ON the node.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.

File WLCOMPort NetworkConfig ChangeDevID Help

WSN-WSN_Main
ResetNW NWDevList Select Device
ResetNW GetDevList FFFF
GetWSN_ID PingDev Time mS
GetWSN_ID PingDev :8000t

WSN RADIO_PARAMETERS
LQIth(1-255) Power in dB Retires
15 4 3
Channel SetRadioPara
3 3 SetRadio

WSN EXPERIMENT PARAMETERS
PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
100 20 20 0
PacketType AckType StartExp Set Expt Par
S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS
StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS
PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication
WiFi:802.11.COM:WiFi AP Clear Screen
saFltDevList.allow_disallow_0:A
saFltDevList:
[*] [Tx]:2,FFFF,8000,0,\$
:2,1008,01,01,205,8000,00,207,\$
[*]

WSN Read/write Sensor I/O
Select dio dio_1 READ_IO READ_IOCont
Config_IO output Temp C Humid %RH
Set_IO hi 0 0
dio_1 Val 0 Press kPa ADC Val
dio_2 Val 0 0 0

WSN Network Parameter and processing
ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

File WLCOMPort NetworkConfig ChangeDevID Help

WSN-WSN_Main
ResetNW NWDevList Select Device
ResetNW GetDevList FFFF
GetWSN_ID PingDev Time mS
GetWSN_ID PingDev :8000t

WSN RADIO_PARAMETERS
LQIth(1-255) Power in dB Retires
15 4 3
Channel SetRadioPara
3 3 SetRadio

WSN EXPERIMENT PARAMETERS
PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
100 20 20 0
PacketType AckType StartExp Set Expt Par
S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS
StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS
PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication
WiFi:802.11.COM:WiFi AP Clear Screen
saFltDevList.allow_disallow_0:A
saFltDevList:
[*] [Tx]:2,FFFF,8000,0,\$
:2,1008,01,01,205,8000,00,207,\$
[*]

WSN Read/write Sensor I/O
Select dio dio_1 READ_IO READ_IOCont
Config_IO output Temp C Humid %RH
Set_IO hi 0 0
dio_1 Val 0 Press kPa ADC Val
dio_2 Val 0 0 0

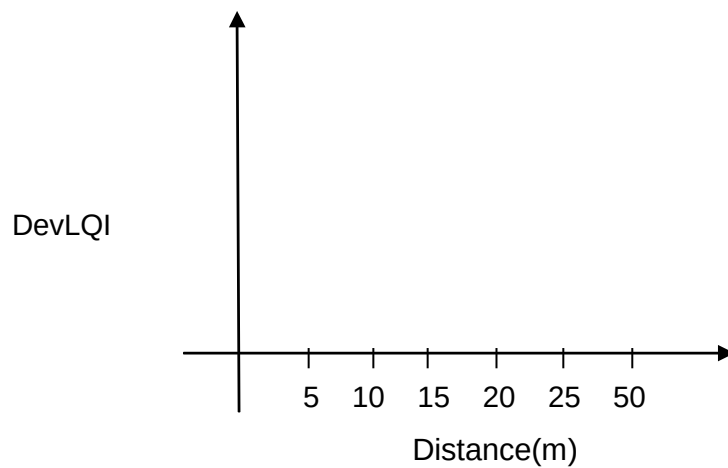
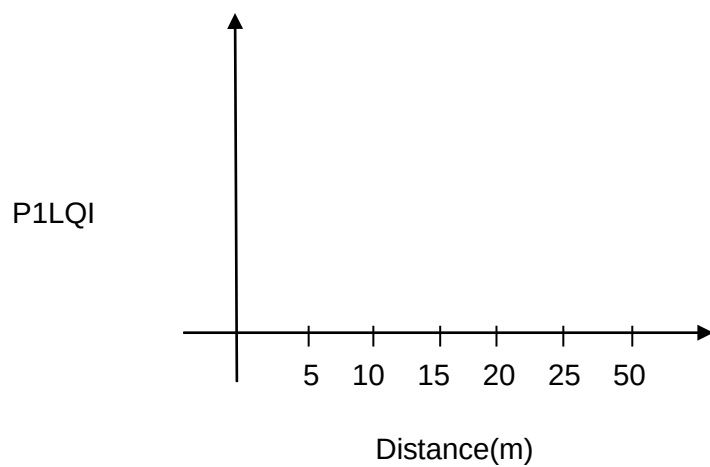
WSN Network Parameter and processing
ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen
Network Device List
Dev |Parents|Hops |DevLQI |Parent1|P1Hop |P1LQI |Parent2|P2Hop |P2LQI
1008|01 |01 |205 |8000 |00 |207 |---- |-- |---

- Check the DevLQI and P1LQI on **WSN Network Parameter and Processing** window
Where, DevLQI is (master to Node Link) and P1LQI is (node to Master Link)
- Now ,measure the LQI at different distance and after each distance press reset button to get the LQI at that distance.
- .

OBSERVATIONS: -

Sr. No.	Distance (in meter)	Master to Node Link LQI(DevLQI)	Node to Master Link LQI(P1LQI)
1.	5 meters		
2.	10 meters		
3.	15 meters		
4.	20 meters		
5.	25 meters		
6.	30 meters		

Now Plot the Graph of LQI v/s Distance.

1. Master to Node Link**2. Node to Master Link LQI(P1LQI)**

Conclusion – As you increase the distance between the master and node the LQI decreases.

The Things affecting the LQI are ,

- Signal Attenuation.
- Increased inference.
- Propagation delay.
- Transmission power Limit.

EXPERIMENT NO. 4

EXPERIMENT NO: 4

NAME: -

Measurement of Throughput Rate and Bandwidth Utilization.

OBJECTIVE: -

- Measurement of Throughput Rate.
- Measurement of bandwidth utilization.

EQUIPMENTS: -

- WL-WSN NPD's 6 to 10 no's.
- Master / Gateway Trainer Kit 01 no
- PC or Laptop with WSN Application software.

THEORY: -

What is ACK?

In some digital communication protocols, ACK is the name of a signal that data has been received successfully (for example, with an acceptable number of errors). The ACK signal is sent by the receiving station (destination) back to the sending station (source) after the receipt of a recognisable block of data of specific size. In order to be recognisable, the data block must conform to the protocol in use. When the source receives the ACK signal from the destination, it transmits the next block of data. If the source fails to receive the ACK signal, it either repeats the block of data or else ceases transmission, depending on the protocol.

No ack

No-Acknowledgement refers to the acknowledge policy used at the MAC level. Enabling no-acknowledgement can result in more efficient Throughput but higher error rates in a noisy Radio Frequency (RF) environment. To disable WMM, deselect Wi-Fi Multimedia.

Hardware Ack (Radio ack)

The benefits of the acknowledgement feature allow a transmitter user the ability to know for sure if a paired receiver has received your transmission signal, even from kilometers away. The green signal confirms that the receiver has received an acknowledgement request from the receiver.

Software ack

Permission is granted to anyone to use this software for any purpose, including commercial applications, and to alter it and redistribute it freely.

Bandwidth

In computer network, **bandwidth** is defined as the maximum number of bits that can flow through a network connection in a given period of time. The fundamental unit of network bandwidth is bits per second (bps).

ISP Bandwidth= 250kbps

Theoretically Bandwidth

250Kbps / 8 = 31,250KBps

Bandwidth = $31 \times 10^3 / (\text{PSByte} + 20 \text{ (Header File Byte)})$

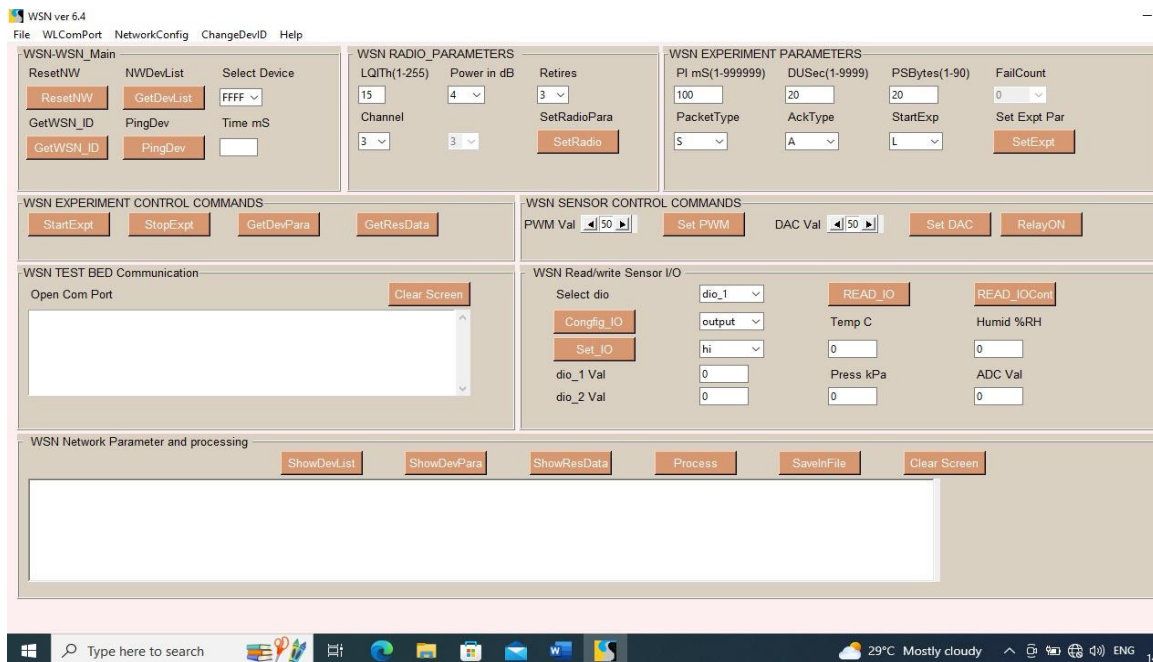
Throughput

In computer network, **Throughput** is defined as the actual number of bits that flows through a network connection in a given period of time. Throughput is always less than or equal to bandwidth but can never exceed bandwidth. In a computer network, the Throughput can be affected by many factors as listed below:

- Network congestion due to heavy network usage.
- Too many users are accessing the same server.
- Low bandwidth allocation between network devices.
- Medium loss of a computer network.
- Resources (CPU, RAM) of network devices.

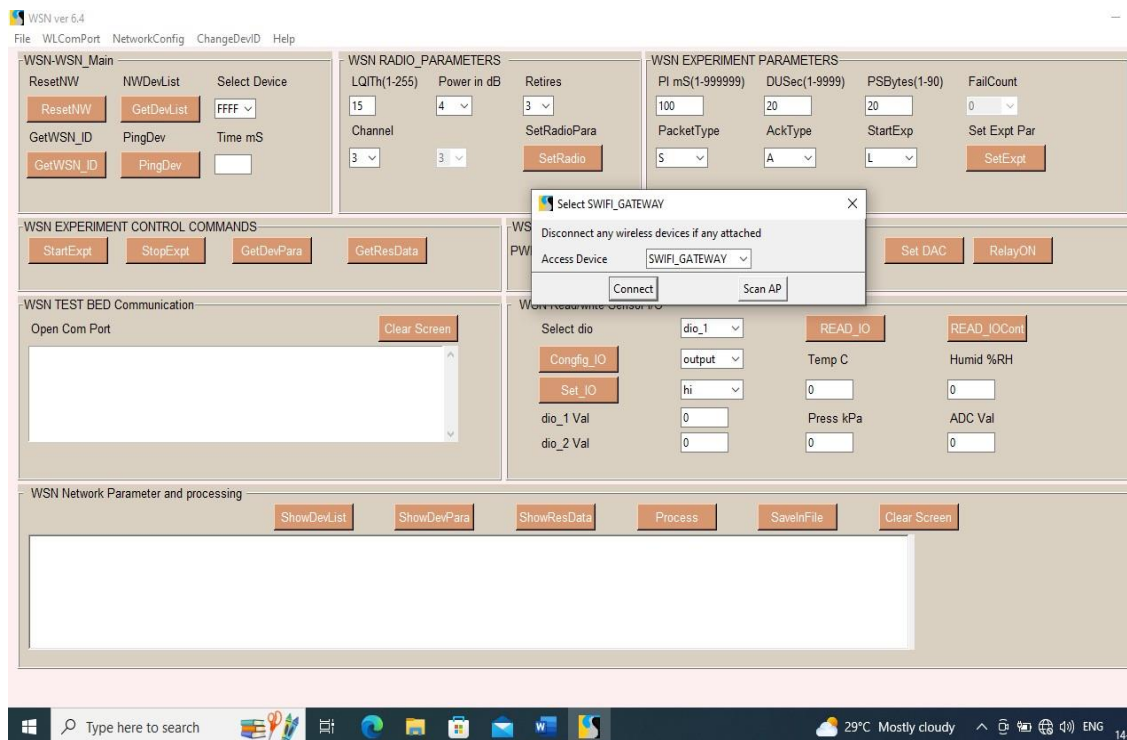
PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.

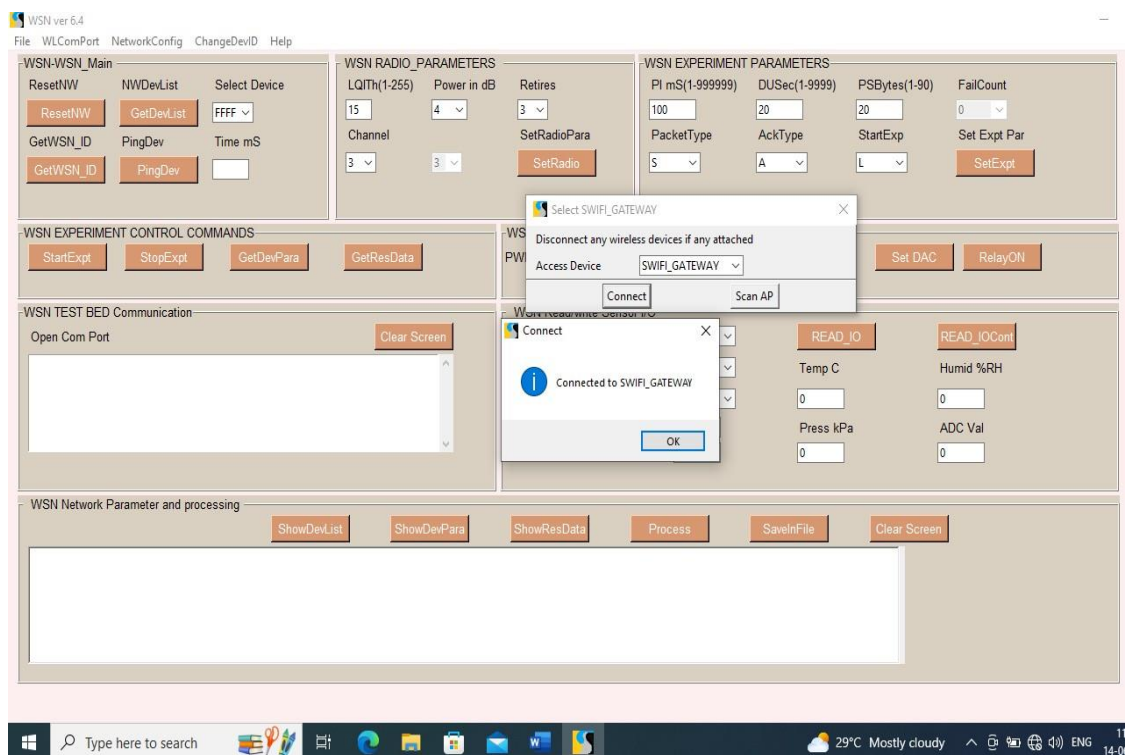


- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.

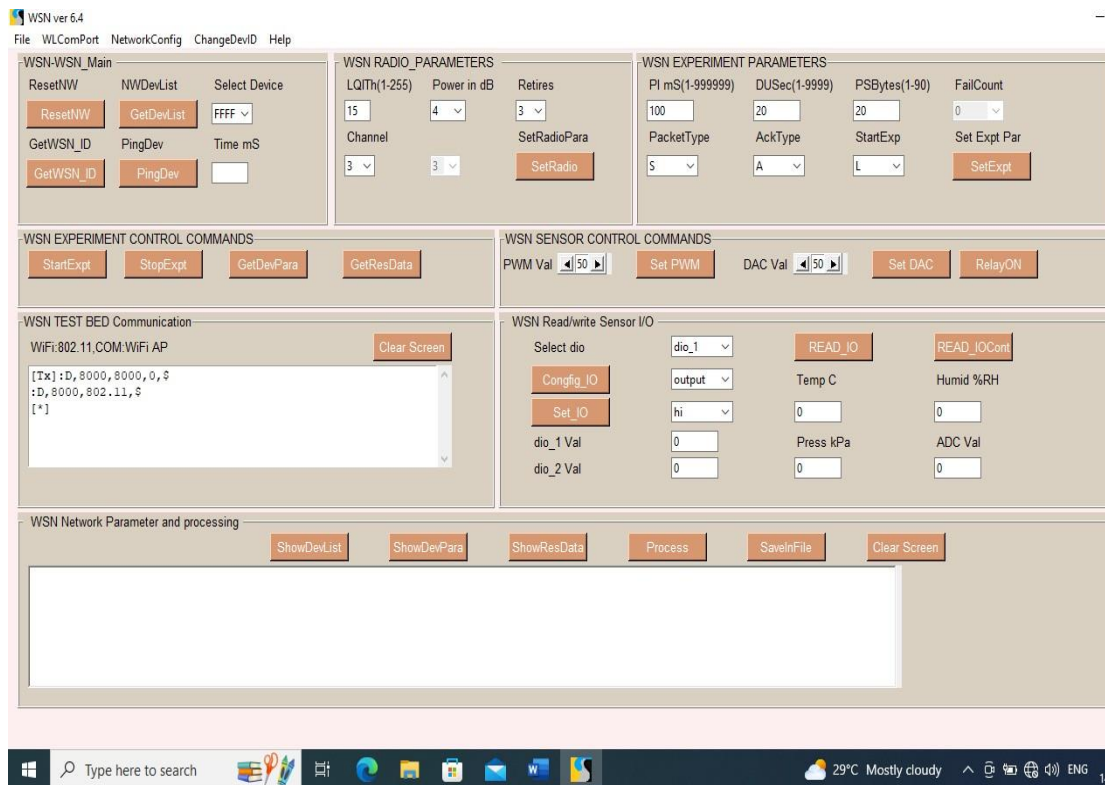
- Now click on WLcomport and select SWIFI_GATEWAY.



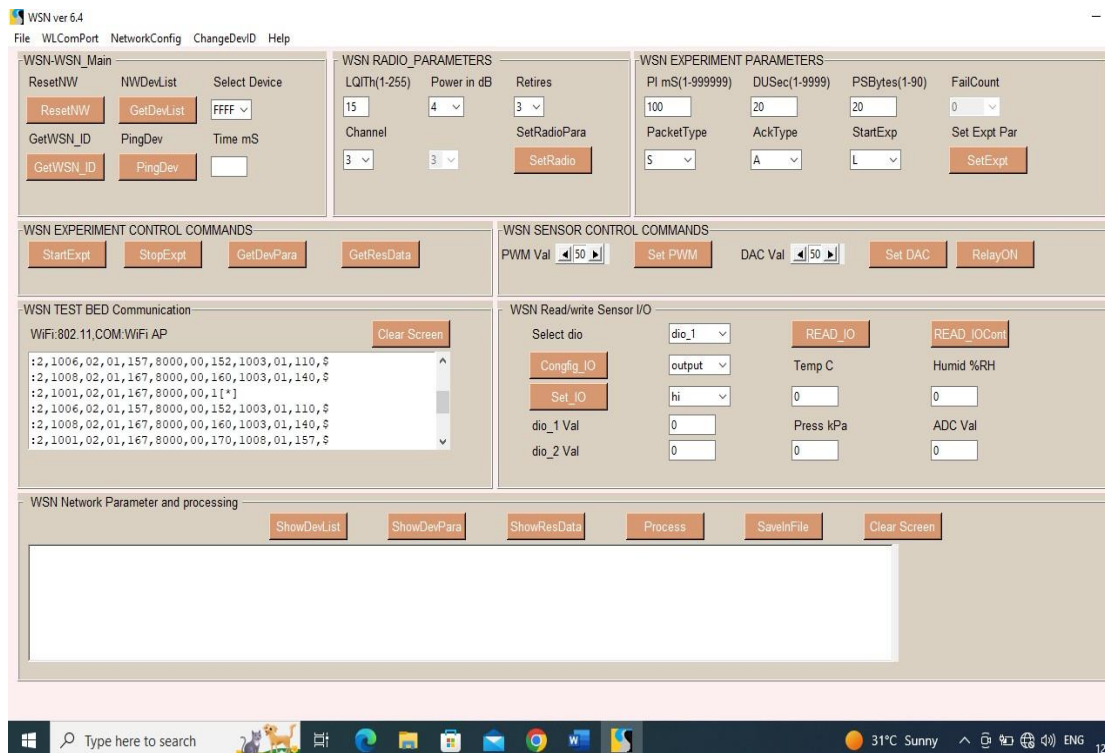
- Click on **Connect** tab.

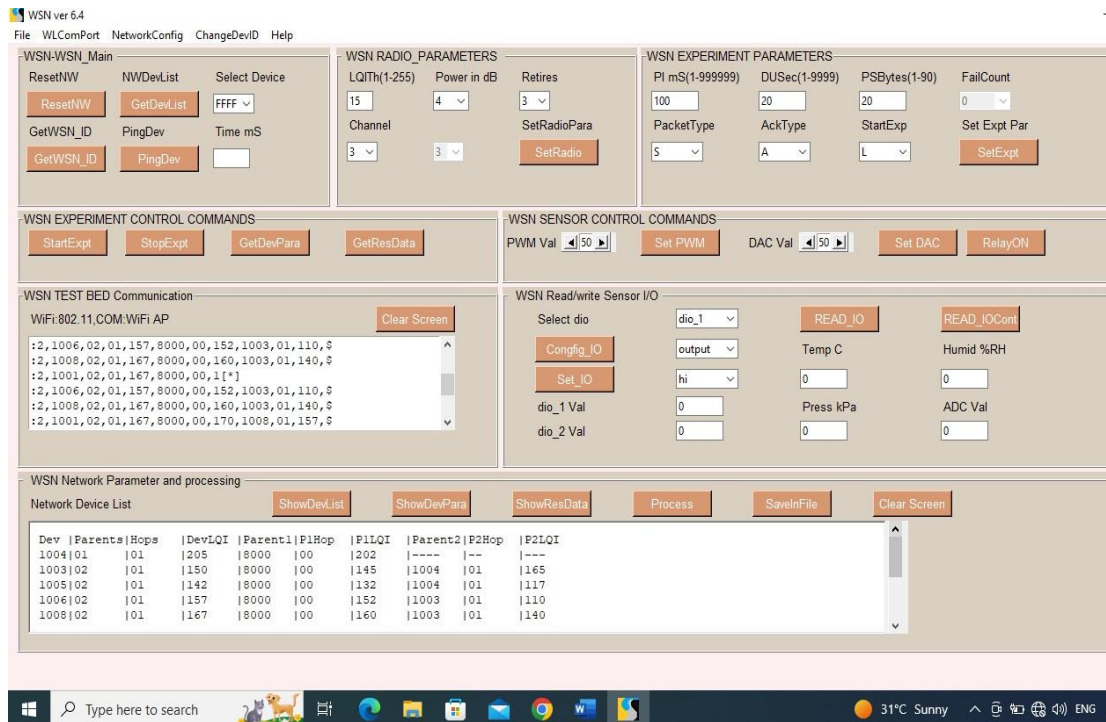


- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.



- Now switch on the all the nodes.
- Reset the network by clicking on ResetNW and check on masters display whether all the nodes are connected or not.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.





- To perform this experiment set the parameter as below observation table.
- Now click on SetExp.
- Then click on the GetDevPara on the (WSN EXPERIMENT CONTROL COMMAND)
- Now Click on the ShowDevPara (WSN Wireless Parameter and Processing). Here we can now see the display of all the device parameters that we have set up for experimentation.
- Then click on the StartExpt. (WSN EXPERIMENT CONTROL COMMAND)
- Now the packet communication from the nodes will start and end after the specified time. (in this case 20 sec.)
- Wait for few sec and then, click on GetResData. (WSN EXPERIMENT CONTROL COMMAND)
- Then click on the ShowResData. (WSN Wireless Parameter and Processing). Here we can now see the value of RxPkt, TXPkt:

Sr. no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission	Throughput rate	Bandwidth utilization
1.	100 ms	20	20	S	N	L		
2.	100 ms	20	20	S	H	L		
3.	100 ms	20	20	S	A	L		
4.	100 ms	20	20	S	R	L		

After getting the values of RxPkt, TxPkt

Calculate RxPkt Bytes and TxPkt Bytes

$\text{RxPkt Bytes} = \text{RxPkt} \times [((\text{Packet size } P_s) + 20 \text{ Bytes (Header)})]$

$\text{TxPkt Bytes} = \text{TxPkt} \times [5 + 20 \text{ Bytes (Header)}]$

$\text{Control Packets} = [5 + 20 \text{ Bytes (header)}]$

Now we calculate Throughput Rate

$\text{Throughput Rate} = (\text{RxPkt Byte} + \text{TxPkt Bytes}) / \text{DU Bytes / sec}$

$\text{Bandwidth utilization} = \text{Actual bandwidth of network} / \text{Throughput Rate Actual}$

band width of network = 31250 KBps

Conclusion-

Here we measure the Throughput Rate for different conditions of the network by changing PI and ack type. In summary, the experiment revealed a direct correlation between acknowledgment mechanisms and throughput in the Wireless Sensor Network. Software acknowledgment emerged as a favorable choice, balancing reliability and efficiency. Hardware acknowledgment, while robust, incurred noticeable overhead. Node-to-node acknowledgment showcased promise in specific architectures. The findings underscore the critical role of acknowledgment strategies in optimizing throughput for varied WSN applications.

EXPERIMENT NO. 5

EXPERIMENT NO: 5

NAME: -

To study Effect of Hops Over Network Transmission Delay for a Tree topology.

OBJECTIVE: -

- Setup End Nodes attached to Master directly.
- Setup End Nodes attached to Master through Single Hop.
- Setup End Nodes attached to Master through Two Hops.
- Configure a Tree topology for multiple Hops network and measurement of network transmission delay with respect to No. of nodes.

EQUIPMENTS: -

- WL-WSN NPD 4 no's
- Master / Gateway Trainer Kit 01 no
- PC or Laptop with WSN Application software.

THEORY: -

What does Hop means?

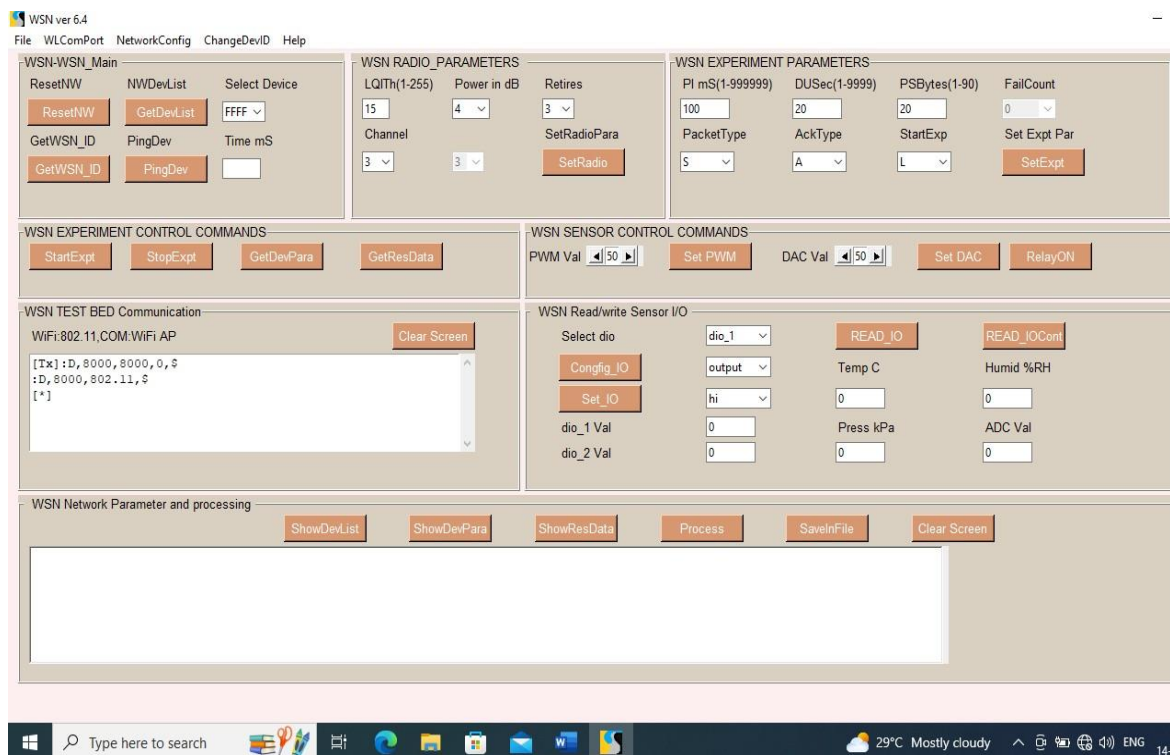
In computer networks, a hop is an intermediate connection in a string of connections linking two devices. Whenever a router or gateway is an intermediary device between two different and remote hosts, nodes or networks, it is known as a hop.

For example, on the Internet, most data packets need to go through several routers before they reach their final destination. Each time the packet is forwarded to the next router, a hop occurs. The more hops, the longer it takes for data to go from source to destination.

All computer networks are composed of many different nodes. The routing of data among these networks is performed using routing logic via a router. A router not only performs routing of data in toward a network but also maintains information about learned paths of different networks.

PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.
- Click on **Connect** tab.
- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.



- Now switch the all the nodes.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.

File WLCOMPort NetworkConfig ChangeDevID Help

WSN-WSN_Main
ResetNW NWDDevList Select Device
ResetNW GetDevList FFFF
GetWSN_ID PingDev Time mS
GetWSN_ID PingDev :8000t

WSN RADIO_PARAMETERS
LQIth(1-255) Power in dB Retires
15 4 3
Channel SetRadioPara
3 3 SetRadio

WSN EXPERIMENT PARAMETERS
PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
100 20 20 0
PacketType AckType StartExp Set Expt Par
5 A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS
StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS
PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication
WiFi:802.11.COM:WiFi AP Clear Screen

```
[*] [Tx]:2,FFFF,8000,0,$
:2,1007,01,01,202,8000,00,185,$
:2,1003,02,01,185,8000,00,185,1007,01,212,$
:2,1008,02,01,215,8000,00,205,1007,01,232,$
:2,1005,02,01,195,8000,00,190,1003,01,182,$
[*]
```

WSN Read/write Sensor I/O
Select dio dio_1 READ_IO READ_IOCont
Config_IO output Temp C Humid %RH
Set_IO hi 0 0
dio_1 Val 0 Press kPa ADC Val
dio_2 Val 0 0

WSN Network Parameter and processing
Network Device List ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
1007 01	01	01	202	8000	00	185	----	---	----
1003 02	01	01	185	8000	00	185	1007	01	212
1008 02	01	01	215	8000	00	205	1007	01	232
1005 02	01	01	195	8000	00	190	1003	01	182

- But first we need to reset the network .
- Click on **NetworkConfig**.

Resetting the network Configuration.

- Now type master address '8000' in **SrcDev** (Source Device).
- Then click on **GetFitList** (Get Filter List).

File WLCOMPort NetworkConfig ChangeDevID Help

WSN-WSN_Main
ResetNW NWDDevList Select Device
ResetNW GetDevList FFFF
GetWSN_ID PingDev Time mS
GetWSN_ID PingDev :8000t

WSN RADIO_PARAMETERS
LQIth(1-255) Power in dB Retires
15 4 3
Channel SetRadioPara
3 3 SetRadio

WSN EXPERIMENT PARAMETERS
PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
100 20 20 0
PacketType AckType StartExp Set Expt Par
5 A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS
StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS
PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication
WiFi:802.11.COM:WiFi AP Clear Screen

```
:2,1003,02,01,185,8000,00,185,1007,01,212,$
:2,1008,02,01,215,8000,00,205,1007,01,232,$
:2,1005,02,01,195,8000,00,190,1003,01,182,$
[*] [Tx]:H,8000,8000,0,$
:H,8000,NUM,00,FLT,Y,A/D,A,FDev,, $
[*]
```

WSN Network Parameter and processing
Network Device List ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
1007 01	01	01	202	8000	00	185	----	---	----
1003 02	01	01	185	8000	00	185	1007	01	212
1008 02	01	01	215	8000	00	205	1007	01	232
1005 02	01	01	195	8000	00	190	1003	01	182

NetworkConfigSetting
SetGet FLT DevList
SrcDev 8000 NoDev 00
Dev1 Dev2
Dev3 Dev4
Dev5 Dev6
Dev7 Dev8
Dev9 Dev10
GetFitList SetList
GetFitList SetList
Control FLT DevList
Apply Allow/Block
Y A
FitListControl Exit
SetFitControl Exit

WSN EXPERIMENT PARAMETERS
PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
100 20 20 0
PacketType AckType StartExp Set Expt Par
5 A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS
StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS
PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication
WiFi:802.11.COM:WiFi AP Clear Screen

```
:2,1003,02,01,185,8000,00,185,1007,01,212,$
:2,1008,02,01,215,8000,00,205,1007,01,232,$
:2,1005,02,01,195,8000,00,190,1003,01,182,$
[*] [Tx]:H,8000,8000,0,$
:H,8000,NUM,00,FLT,Y,A/D,A,FDev,, $
[*]
```

WSN Network Parameter and processing
Network Device List ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
1007 01	01	01	202	8000	00	185	----	---	----
1003 02	01	01	185	8000	00	185	1007	01	212
1008 02	01	01	215	8000	00	205	1007	01	232
1005 02	01	01	195	8000	00	190	1003	01	182

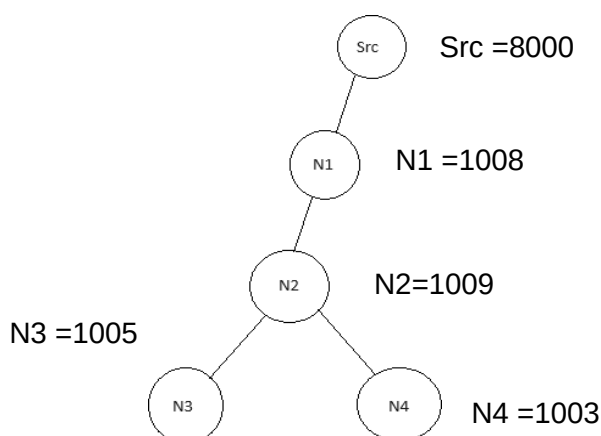
- After click on get filter list, we will get the previous setting of network configuration connected to Src Device.

- Now delete all the data from device list if present in (Dev1,Dev2,Dev3,..) except source address and then click on **SetList**.
- Also if Write 00 in NoDev (number of device connected)
- Now select apply as 'Y' and Allow/Block as 'A'.
- Then click on **FltListControl**. (This command is for set new configuration.)
- Now **ResetNw**. This command will reset the network and reset on default setting.
- Now click **GetDevList**. We will get the current network status.
- Similarly check for nodes too just write the address of the device and click on **GetFltList**, if see any device connected repeat the same process which we did for master.
- Observe the setting on Show Device List.

Now we will create tree topology as per the diagram given below given below.

(For eg. Src = 8000, N1=1008, N2=1009, N3=1005, N4=1003)

Note- these address are just an example to understand, enter the address of the nodes which are present in your network.



- Click on **NetworkConfig**.
- Now type Master address(8000) in source address, select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now let's create hops in the network, type source address as '8000' and type detected N1 device address in **Dev1**.
- Now type **NoDev** as '1'.
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now reset the network and check hop is created or not.
- Now again click on **NetworkConfig**.
- Now select **Dev1** (N1) address as source address and type another detected NPD (N2) as Dev1.
- Now type **NoDev** as '1'
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- This step will create another hop in network.
- Now select **Dev1** (N2) address as source address and type another detected NPD (N3) as Dev1 and another NPD (N4).

- Now type **NoDev** as '2'
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control. This step will create another hop in network and we get the Tree Topology .

The screenshot displays the WSN configuration software interface with the following sections:

- WSN-WSN_Main:** Includes buttons for ResetNW, GetDevList, Select Device, GetWSN_ID, PingDev, and Time mS.
- WSN RADIO PARAMETERS:** Includes LQIth(1-255), Power in dB, Retires, Channel, and SetRadioPara.
- WSN EXPERIMENT PARAMETERS:** Includes Pl mS(1-999999), DUSec(1-9999), PSBytes(1-90), FailCount, PacketType, AckType, StartExp, and Set Expt Par.
- WSN EXPERIMENT CONTROL COMMANDS:** Includes StartExpt, StopExpt, GetDevPara, and GetResData.
- WSN SENSOR CONTROL COMMANDS:** Includes PWM Val, Set PWM, DAC Val, Set DAC, and RelayON.
- WSN TEST BED Communication:** Includes a text area for WiFi: 802.11.COM:WiFi AP and a Clear Screen button.
- WSN Read/write Sensor I/O:** Includes Select dio, Config_IO, Set_IO, READ_IO, READ_IOCont, Temp C, Humid %RH, Press kPa, and ADC Val.
- WSN Network Parameter and processing:** Includes buttons for ShowDevList, ShowDevPara, ShowResData, Process, SaveInFile, and Clear Screen.

The Network Device List table is shown below:

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
1007	01	01	200	8000	100	190	----	--	----
1003	01	02	195	1007	01	220	----	--	----
1008	02	02	210	1007	01	237	1003	02	195
1005	02	02	155	1007	01	155	1003	02	177

Now, set these parameter and perform the experiment just as experiment and get the value of TxATm(Avg transmission time)

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L

Conclusion-

- In this experiment we learned how to create hops in the network and how we can make different topologies .
- As the hops increases in tree topology we notice a transmission delay over the network.

EXPERIMENT NO. 6

EXPERIMENT NO: 6

NAME: -

To study the Effect of Packet Payload on Network

OBJECTIVE: -

Study the Effect of Packet Payload on Network Throughput.

EQUIPMENTS: -

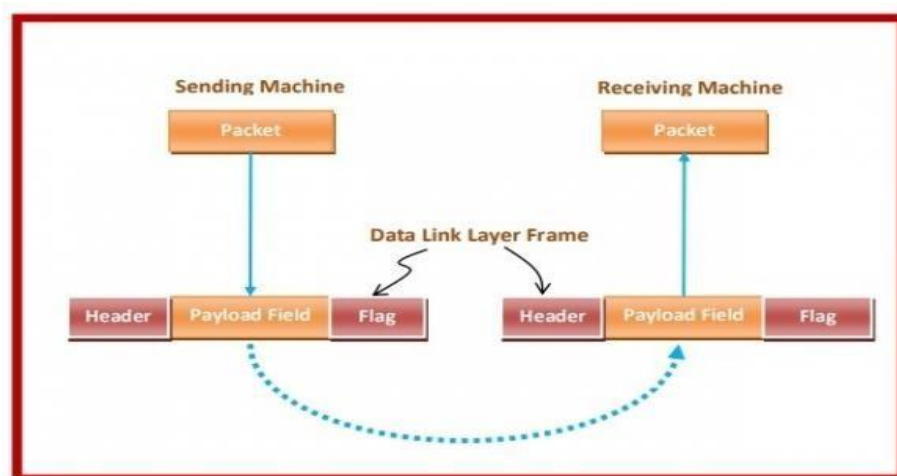
- WL-WSN NPD's 10 no's
- Master / Gateway Trainer 01 no
- PC or Laptop with GNU Application software.

THEORY: -

What is PAYLOAD in networking?

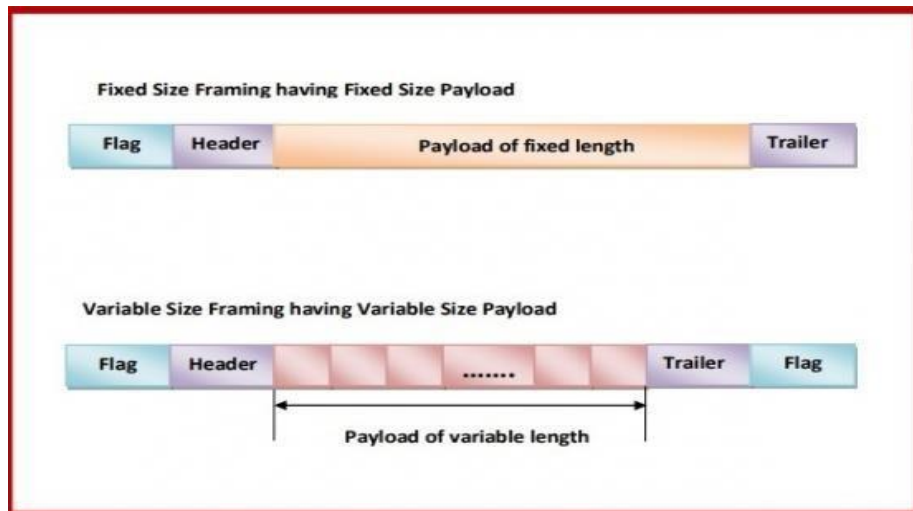
In computer networking and telecommunications, when a transmission unit is sent from the source to the destination, it contains both a header and the actual data to be transmitted. This actual data is called the payload. The header contains the protocol information as well as the source and destination addresses, which are required for delivery of the payload. The payload is required by the destination client.

The following diagram depicts encapsulation of payload field in a data link layer frame. The data packet from the upper layer is framed by adding a header that contains the control information for data transmission, the payload field containing the actual data and the end flag that marks the end of the data frame.



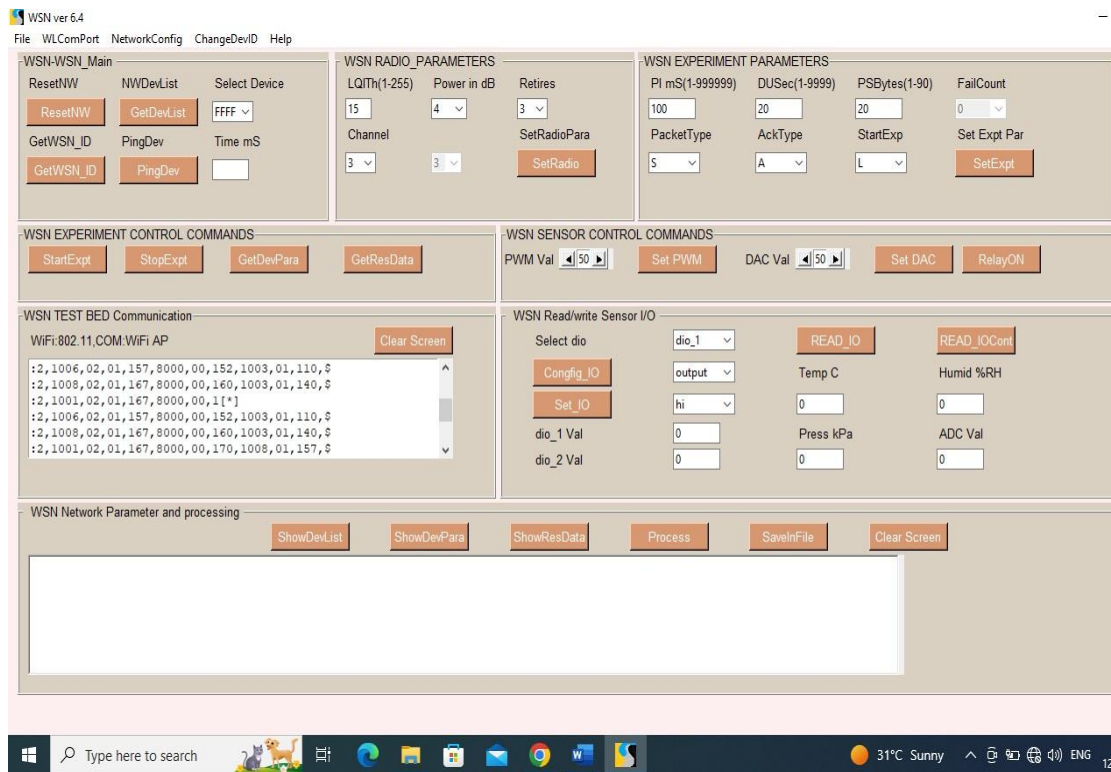
Fixed and Variable Payload

The payload field may be of fixed size or of variable size. If the payload field is of fixed size, then the frame becomes of fixed size. Fixed size frames do not need any end flag, since the frame length acts as delimiter of the frame. In case of variable length payload, a pattern is used as a delimiter to determine the size of frame. Alternatively, a length field is kept that contains the frame size.

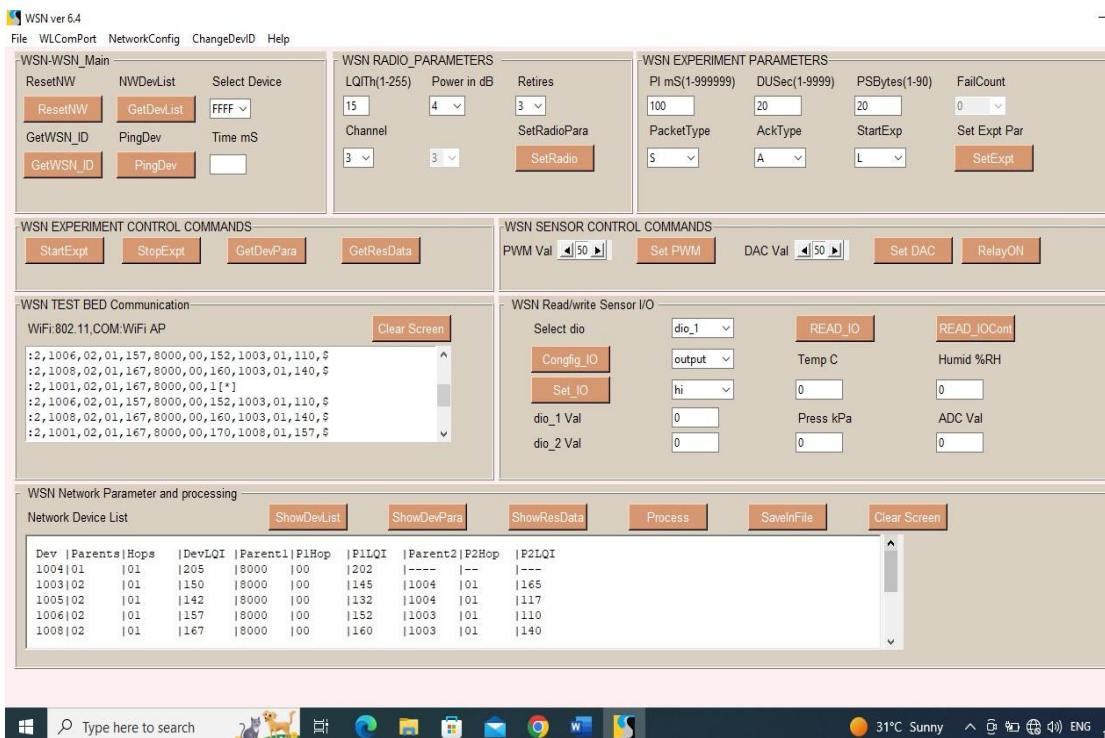


PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.
- Click on the Connect tab.
- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.
- Now switch on the nodes.



- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.



- For this experiment we have to vary the parameter from **WSN EXPERIMENT PARAMETER**.
- Set the parameter as below and start the experiment.

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission	Throughput rate
1.	100 ms	20	10	S	A	L	
2.	100 ms	20	20	S	A	L	
3.	100 ms	20	30	S	A	L	
4.	100 ms	20	40	S	A	L	
5.	100 ms	20	50	S	A	L	

After getting the values of RxPkt, TxPkt

Calculate RxPkt bytes and TxPkt

$\text{RxPkt Bytes} = \text{RxPkt} \times [((\text{Packet size Ps}) + 20 \text{ Bytes (Header)})]$

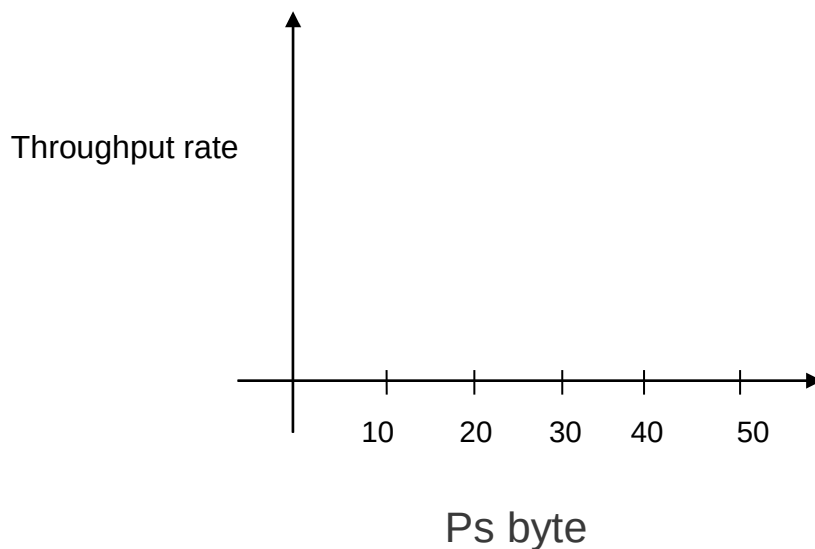
$\text{TxPkt Bytes} = \text{TxPkt} \times [5 + 20 \text{ Bytes (Header)}]$

$\text{Control Packets} = [(5 + 20 \text{ Bytes (header)})]$

Now we calculate Throughput Rate

$\text{Throughput Rate} = (\text{RxPkt Byte} + \text{TxPkt Bytes}) / \text{DU Bytes / sec}$

Now Plot the Graph of Throughput rate v/s PSByte.



Conclusion –

As we increase the size of data packets we get better Throughput

EXPERIMENT NO. 7

EXPERIMENT NO: 7

NAME: -

To study Effect of Packet Transmission Rate Over Network .

OBJECTIVE: -

Study Effect of Packet Transmission Rate on Network .

EQUIPMENTS: -

- WL-WSN NPD 10 no's
- Master / Gateway Trainer Kit 01 no
- PC or Laptop with GUI WSN Application software.

PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.
- Click on the Connect tab.
- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.
- Now switch on the nodes.
- And reset the network by clicking on ResetNw.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.

WSN ver 6.4
File WLComPort NetworkConfig ChangeDevID Help

WSN-WSN_Main

ResetNW NWDevList Select Device

ResetNW GetDevList FFFF

GetWSN_ID PingDev Time mS

GetWSN_ID PingDev

WSN RADIO PARAMETERS

LQIth(1-255) Power in dB Retires

15 4 3

Channel SetRadioPara

3 3 SetRadio

WSN EXPERIMENT PARAMETERS

PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount

100 20 20 0

PacketType AckType StartExp Set Expt Par

S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS

StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS

PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication

WiFi:802.11.COM:WiFi AP Clear Screen

```

:2,1006,02,01,157,8000,00,152,1003,01,110,$
:2,1008,02,01,167,8000,00,160,1003,01,140,$
:2,1001,02,01,167,8000,00,1[*]
:2,1006,02,01,157,8000,00,152,1003,01,110,$
:2,1008,02,01,167,8000,00,160,1003,01,140,$
:2,1001,02,01,167,8000,00,170,1008,01,157,$

```

WSN Read/write Sensor I/O

Select dio dio_1 READ_IO READ_IOCont

Config_IO output Temp C Humid %RH

Set_IO hi 0 0

dio_1 Val 0 Press kPa ADC Val

dio_2 Val 0 0 0

WSN Network Parameter and processing

ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

WSN ver 6.4
File WLComPort NetworkConfig ChangeDevID Help

WSN-WSN_Main

ResetNW NWDevList Select Device

ResetNW GetDevList FFFF

GetWSN_ID PingDev Time mS

GetWSN_ID PingDev

WSN RADIO PARAMETERS

LQIth(1-255) Power in dB Retires

15 4 3

Channel SetRadioPara

3 3 SetRadio

WSN EXPERIMENT PARAMETERS

PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount

100 20 20 0

PacketType AckType StartExp Set Expt Par

S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS

StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS

PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication

WiFi:802.11.COM:WiFi AP Clear Screen

```

:2,1006,02,01,157,8000,00,152,1003,01,110,$
:2,1008,02,01,167,8000,00,160,1003,01,140,$
:2,1001,02,01,167,8000,00,1[*]
:2,1006,02,01,157,8000,00,152,1003,01,110,$
:2,1008,02,01,167,8000,00,160,1003,01,140,$
:2,1001,02,01,167,8000,00,170,1008,01,157,$

```

WSN Read/write Sensor I/O

Select dio dio_1 READ_IO READ_IOCont

Config_IO output Temp C Humid %RH

Set_IO hi 0 0

dio_1 Val 0 Press kPa ADC Val

dio_2 Val 0 0 0

WSN Network Parameter and processing

ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

Network Device List

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
1004 01	01	205	8000	00	202	----	--	---	---
1003 02	01	150	8000	00	145	1004	01	165	117
1005 02	01	142	8000	00	132	1004	01	117	110
1006 02	01	157	8000	00	152	1003	01	110	140
1008 02	01	167	8000	00	160	1003	01	140	140

- For this experiment we have to vary the parameter from **WSN EXPERIMENT PARAMETER**.

- Set the parameter as above.

OBSERVATION: -

Sr.No.	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission	Throughput rate
1.	50 ms	20	20	S	A	L	
2.	100 ms	20	20	S	A	L	
3.	200 ms	20	20	S	A	L	
4.	300 ms	20	20	S	A	L	
5.	400 ms	20	20	S	A	L	
6.	500 ms	20	20	S	A	L	

After getting the values of RxPkt, TxPkt

Calculate RxPkt bytes and TxPkt

$\text{RxPkt Bytes} = \text{RxPkt} \times [((\text{Packet size Ps}) + 20 \text{ Bytes (Header)})]$

$\text{TxPkt Bytes} = \text{TxPkt} \times [5 + 20 \text{ Bytes (Header)}]$

As,

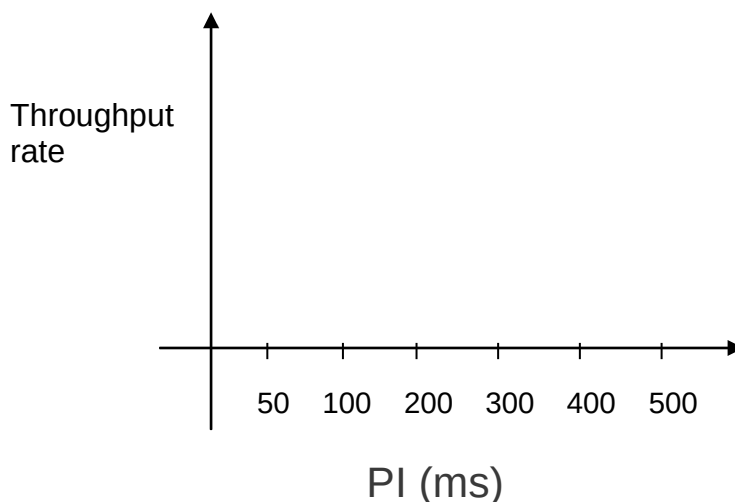
$\text{ControlPackets} = [(5 + 20 \text{ Bytes (header)})]$

Now we calculate Throughput Rate

$\text{Throughput Rate} = (\text{RxPkt Byte} + \text{TxPkt Bytes}) / \text{DU Bytes / sec}$

Graph-

Now Plot the Graph of Throughput v/s PI mS.



Conclusion – As we increase the packet interval the Throughput decreases .A longer packet interval mean that there is more time between the transmission of each packet. Which can lead to a lower rate of data transmission and hence Throughput decrease.

EXPERIMENT NO. 8

EXPERIMENT NO: 8

NAME: -

To study and compare Network Throughput in Outdoor (line of sight) and indoor setup.

OBJECTIVE: -

Compare Network Throughput in Outdoor and Indoor setup.

EQUIPMENTS: -

- WL-WSN NPD 10 no's
- Master / Gateway Trainer 01 no
- PC or Laptop with WSN Application software.

PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.
- Click on the Connect tab.
- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.
- Now switch on all the nodes.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.

WSN ver 6.4
File WLCmPort NetworkConfig ChangeDevID Help

WSN-WSN_Main

ResetNW NWDevList Select Device

ResetNW GetDevList FFFF

GetWSN_ID PingDev Time mS

GetWSN_ID PingDev

WSN RADIO PARAMETERS

LQIth(1-255) Power in dB Retires

15 4 3

Channel SetRadioPara

3 3 SetRadio

WSN EXPERIMENT PARAMETERS

PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount

100 20 20 0

PacketType AckType StartExp Set Expt Par

S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS

StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS

PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication

WiFi:802.11.COM:WiFi AP Clear Screen

```

:2,1006,02,01,157,8000,00,152,1003,01,110,0
:2,1008,02,01,167,8000,00,160,1003,01,140,0
:2,1001,02,01,167,8000,00,160,1003,01,110,0
:2,1006,02,01,157,8000,00,152,1003,01,110,0
:2,1008,02,01,167,8000,00,160,1003,01,140,0
:2,1001,02,01,167,8000,00,170,1008,01,157,0

```

WSN Read/write Sensor I/O

Select dio dio_1 READ_IO READ_IOCont

Config_IO output Temp C Humid %RH

Set_IO hi 0 0

dio_1 Val 0 Press kPa ADC Val

dio_2 Val 0 0 0

WSN Network Parameter and processing

ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

WSN ver 6.4
File WLCmPort NetworkConfig ChangeDevID Help

WSN-WSN_Main

ResetNW NWDevList Select Device

ResetNW GetDevList FFFF

GetWSN_ID PingDev Time mS

GetWSN_ID PingDev

WSN RADIO PARAMETERS

LQIth(1-255) Power in dB Retires

15 4 3

Channel SetRadioPara

3 3 SetRadio

WSN EXPERIMENT PARAMETERS

PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount

100 20 20 0

PacketType AckType StartExp Set Expt Par

S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS

StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS

PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication

WiFi:802.11.COM:WiFi AP Clear Screen

```

:2,1006,02,01,157,8000,00,152,1003,01,110,0
:2,1008,02,01,167,8000,00,160,1003,01,140,0
:2,1001,02,01,167,8000,00,160,1003,01,110,0
:2,1006,02,01,157,8000,00,152,1003,01,110,0
:2,1008,02,01,167,8000,00,160,1003,01,140,0
:2,1001,02,01,167,8000,00,170,1008,01,157,0

```

WSN Read/write Sensor I/O

Select dio dio_1 READ_IO READ_IOCont

Config_IO output Temp C Humid %RH

Set_IO hi 0 0

dio_1 Val 0 Press kPa ADC Val

dio_2 Val 0 0 0

WSN Network Parameter and processing

ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

Network Device List

Dev	Parents	Hops	DevLQI	Parent1	P1Hop	P1LQI	Parent2	P2Hop	P2LQI
1004 01	01	205	8000	00	202	---	---	---	---
1003 02	01	150	8000	00	145	1004	01	165	
1005 02	01	142	8000	00	132	1004	01	117	
1006 02	01	157	8000	00	152	1003	01	110	
1008 02	01	167	8000	00	160	1003	01	140	

- For this experiment we have to vary the parameter from **WSN EXPERIMENT PARAMETER**.

Observation Table –

Sr .n o	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transm ission	Throughput Rate Indoor	Throughpu t Rate Outdoor
1.	100 ms	20	20	S	N	L		
2.	100 ms	20	20	S	A	L		

Set these parameter in indoor and outdoor environment

EXPERIMENT NO. 9

EXPERIMENT NO: 9

NAME: -

Calculation of Communication Load , network efficiency, device efficiency in different topologies like Star, Tree and linear network.

OBJECTIVE: -

- Setup End Nodes attached to Master with Star Topology.
- Setup End Nodes attached to Master through Tree Topology.
- Setup End Nodes attached to Master through Linear Topology.

EQUIPMENTS: -

- WL-WSN NPD 10 no's
- Master / Gateway Trainer Kit 01 no
- PC or Laptop with WSN Application software.

THEORY: -

What does Hop means?

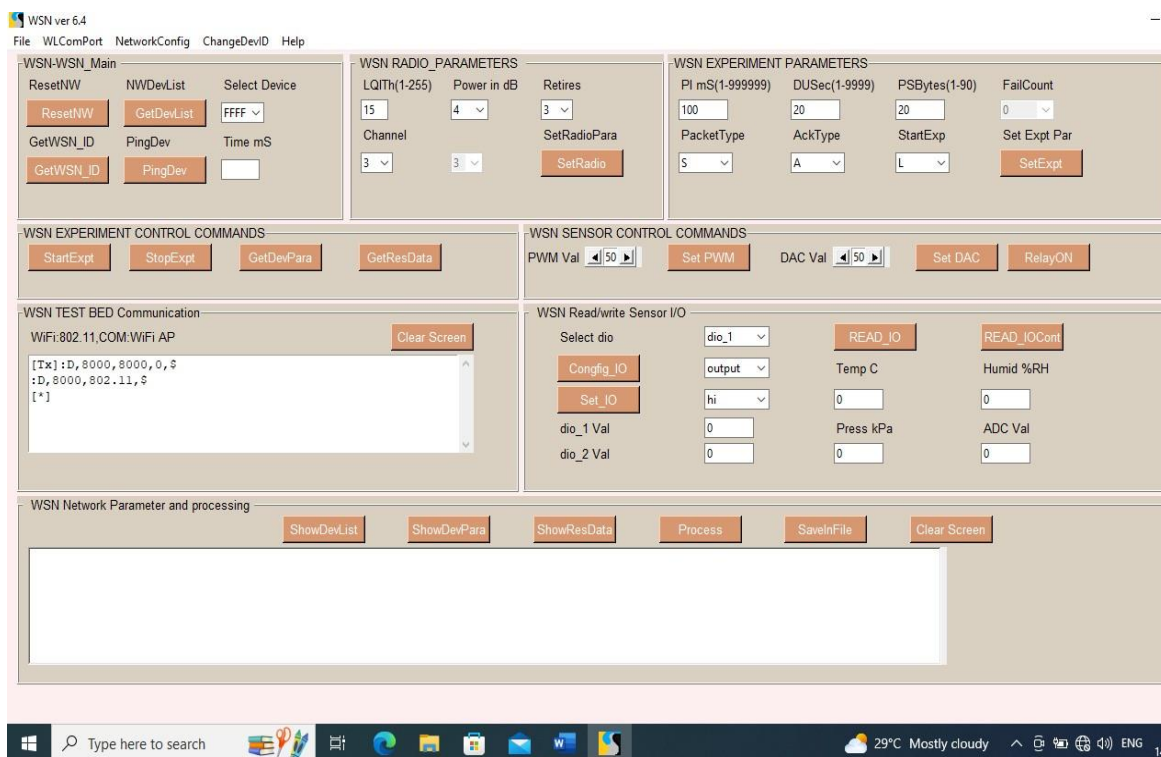
In computer networks, a hop is an intermediate connection in a string of connections linking two devices. Whenever a router or gateway is an intermediary device between two different and remote hosts, nodes or networks, it is known as a hop.

For example, on the Internet, most data packets need to go through several routers before they reach their final destination. Each time the packet is forwarded to the next router, a hop occurs. The more hops, the longer it takes for data to go from source to destination.

All computer networks are composed of many different nodes. The routing of data among these networks is performed using routing logic via a router. A router not only performs routing of data in toward a network but also maintains information about learned paths of different networks.

PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.
- Ensure that you are not connected to any Wi-Fi or network. If you are connected, please disconnect.
- Now click on WLcomport and select SWIFI_GATEWAY.
- Click on **Connect** tab.
- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.



- Now switch the all the nodes.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.

File WLCmPort NetworkConfig ChangeDevID Help

WSN-WSN_Main

ResetNW NWDDevList Select Device
 ResetNW GetDevList FFFF
 GetWSN_ID PingDev Time mS
 GetWSN_ID PingDev :8000t

WSN RADIO PARAMETERS

LQI(1-255) Power in dB Retires
 15 4 3
 Channel SetRadioPara
 3 3 SetRadio

WSN EXPERIMENT PARAMETERS

PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount
 100 20 20 0
 PacketType AckType StartExp Set Expt Par
 S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS

StartExpt StopExpt GetDevPara GetResData

WSN TEST BED Communication

WiFi:802.11.COM:WiFi AP Clear Screen

```

12,1009,02,01,205,8000,00,147,1008,01,210,["*"]
12,1004,02,01,232,8000,00,232,1008,01,180,$
12,1009,02,01,205,8000,00,147,1008,01,210,$
12,1003,02,01,205,8000,00,212,1004,01,195,$
12,1007,02,01,165,8000,00,205,1009,01,152,$
["*"]
  
```

WSN Read/write Sensor I/O

Select dio dio_1 READ_IO READ_IOCet
 Config_IO output Temp C Humid %RH
 Set_IO hi 0 0
 dio_1 Val 0 Press kPa ADC Val
 dio_2 Val 0 0 0

WSN Network Parameter and processing

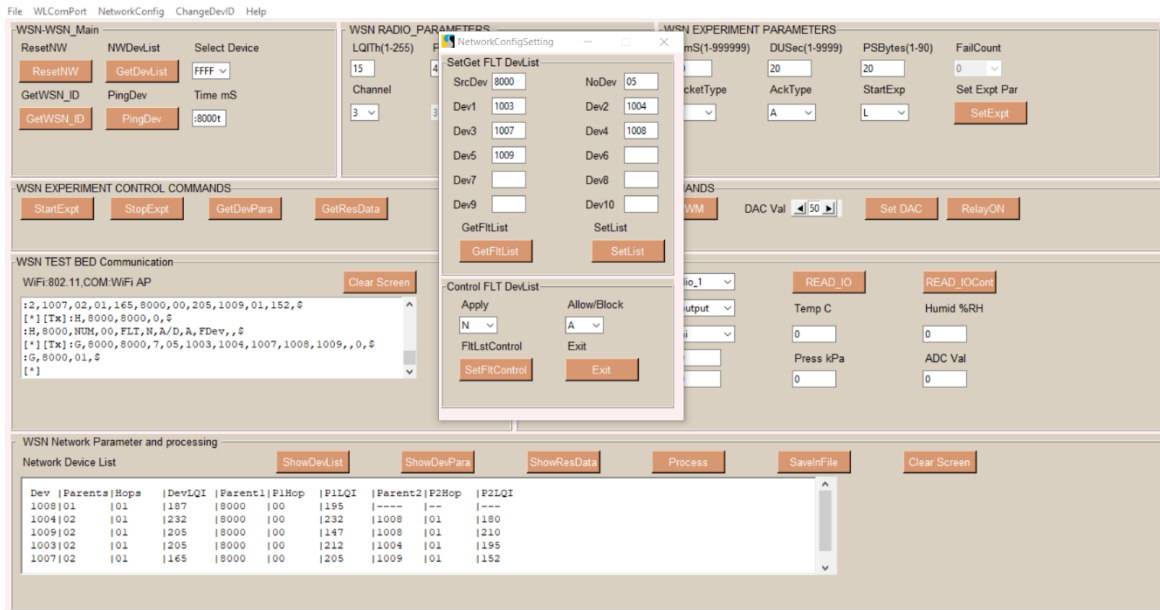
Network Device List ShowDevList ShowDevPara ShowResData Process SaveFile Clear Screen

Dev	Parents	Hops	DevLQI	Parent1	F1Hop	F1LQI	Parent2	F2Hop	F2LQI
1008	01	01	187	8000	100	195	----	---	----
1004	02	01	232	8000	100	232	1008	01	180
1009	02	01	1205	8000	100	1247	1008	01	1210
1003	02	01	1205	8000	100	1212	1004	01	1195
1007	02	01	1165	8000	100	1205	1009	01	1152

- Now we will create numbers of Hops in the Network.
- Click on **NetworkConfig**.

Resetting the network Configuration.

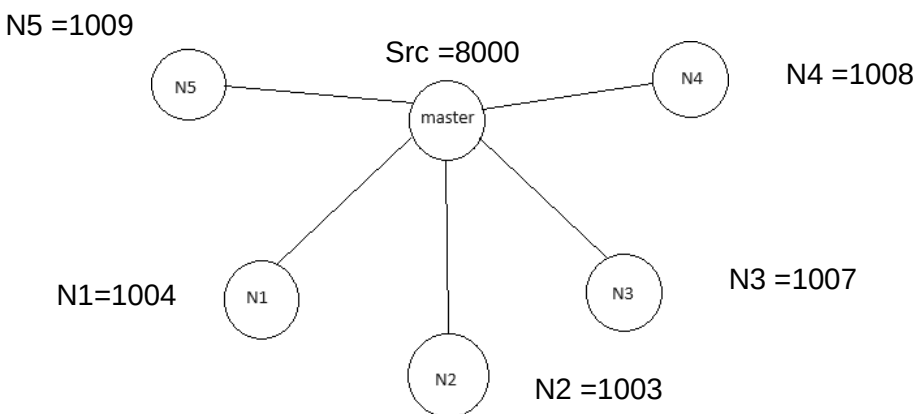
- Now type master address '8000' in **SrcDev** (Source Device).
- Then click on **GetFltList** (Get Filter List).
- After click on get filter list, we will get the previous setting of network configuration connected to Src Device.
- Now delete all the data from device list if present in (Dev1,Dev2,Dev3,...) except source address and then click on **SetList**.
- Also if Write 00 in NoDev (number of device connected)
- Now select apply as 'Y' and Allow/Block as 'A'.
- Then click on **FltListControl**. (This command is for set new configuration.)
- Now **ResetNw**. This command will reset the network and reset on default setting.
- Now click **GetDevList**. We will get the current network status.
- Observed the setting on Show Device List.



- First reset the network as mentioned in **experiment**

Case 1 : Star Topology

Now Lets make a network based on this image given below.



- Click on **NetworkConfig** and set as above.
- Now type Master address(8000) in source address, select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now let's create the network, type source address as '8000' and type detected N1 device address in **Dev1=1003, Dev2=1003, Dev3=1007, Dev4=1008, Dev5=1009**.
- Now type **NoDev** as '5'.
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now reset the network and check hop is created or not.

By setting these parameters perform the experiment as mentioned in experiment No.2

Sr .no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	PacketType	ACT Type	Packet Transmission
1.	50 ms	20	20	S	N	L
2.	100 ms	20	20	S	N	L
3.	150 ms	20	20	S	N	L

Sr .no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	PacketType	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L
2.	100 ms	20	20	S	A	L

As,we have got the ResData on the (WSN Wireless parameter and Processing) Click on **Process** to get Communication Load , network efficiency, device efficiency

File WLComPort NetworkConfig ChangeDevID Help

WSN-WSN_Main

ResetNW NWDvList Select Device

ResetNW GetDevList FFFF

GetWSN_ID PingDev Time mS

GetWSN_ID PingDev 1007

WSN RADIO_PARAMETERS

LQIth(1-255) Power in dB Retires

15 4 3

Channel SetRadioPara

3 3 SetRadio

WSN EXPERIMENT PARAMETERS

PI mS(1-999999) DUSec(1-9999) PSBytes(1-90) FailCount

100 20 20 0

PacketType AckType StartExp Set Expt Par

S A L SetExpt

WSN EXPERIMENT CONTROL COMMANDS

StartExpt StopExpt GetDevPara GetResData

WSN SENSOR CONTROL COMMANDS

PWM Val 50 Set PWM DAC Val 50 Set DAC RelayON

WSN TEST BED Communication

WiFi:802.11.COM:WiFi AP

Clear Screen

53,N,0000,TA,0000:0000:4095,LA,211:227:112,\$

:A,1007,T,0198,R,0201,D,0198,TF,0000,DF,0000,TC,0014,RC,02

20,N,2153,TA,0014:0030:0007,LA,213:222:207,BV,382,AC,0937,

\$

[*]

WSN Read/write Sensor I/O

Select dio dio_1 READ_IO READ_IOCont

Config_IO output Temp C Humid %RH

Set_IO hi 0 0

dio_1 Val 0 Press kPa ADC Val

dio_2 Val 0 0 0

WSN Network Parameter and processing

ShowDevList ShowDevPara ShowResData Process SaveInFile Clear Screen

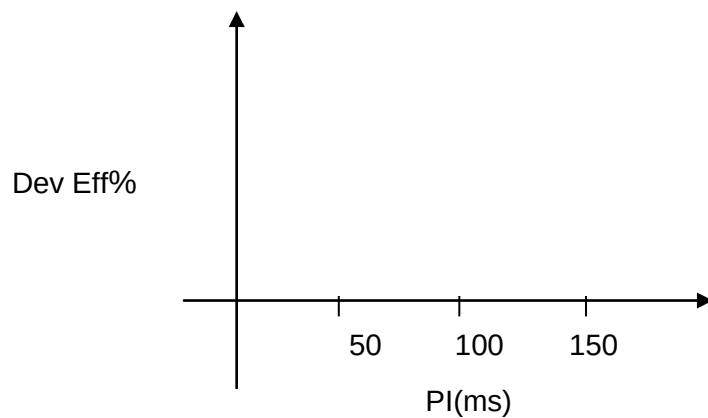
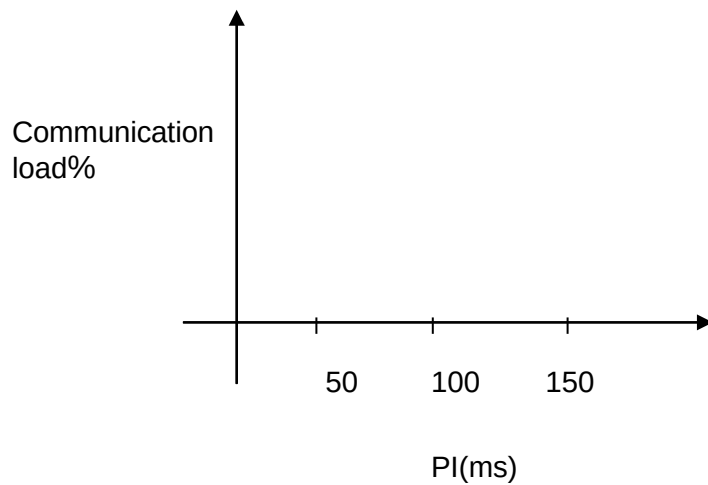
Show Processed Results

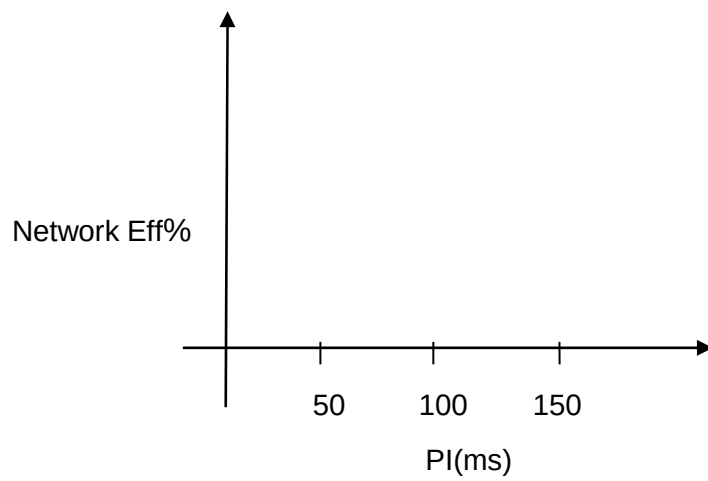
Dev	RxTxDPkt	IDPkt	CommLoad%	Dev Effi%	NwExpEffi%	NwEffi%
8000	0900	0	110.778	0	90.0	100.671
1008	0102	200	140.196	51.0	0	0
1004	0198	200	123.737	99.0	0	0
1003	0198	200	120.707	99.0	0	0
1009	0198	200	120.707	99.0	0	0

Observation Table for Star Topology

PI	50ms	100ms	150ms
Communication Load %			
Device Efficiency%			
Network Efficiency%			

Now we will plot a Graph for Communication Load, Device Efficiency , Network Efficiency Vs PI



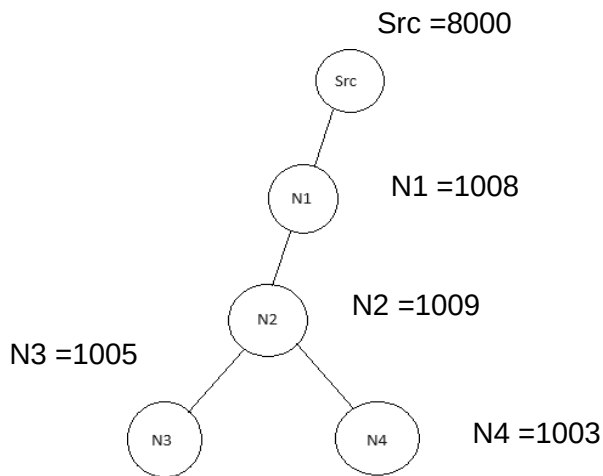


Observation Table for Star Topology

Ack Type	N (No Acknowledgement)	A (Software Acknowledgement)
Communication Load%		
Device Efficiency%		
Network Efficiency%		

Case 2- Tree Topology

- Before we start Tree topology Reset the network configuration



- Click on **NetworkConfig** and set as above.
- Now type Master address(8000) in source address, select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now let's create hops in the network,type source address as '8000' and type detected N1 device address in Eg.**Dev1=1008**.
- Now type **NoDev** as '1'.
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now reset the network and check hop is created or not.
- Now again click on **NetworkConfig**.
- Now select **SrcDev** (N1=1008) address as source address and type another detected NPD (N2=1009) as Dev1.
- Now type **NoDev** as '1'
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- This step will create another hop in network.
- Now select **SrcDev** (Eg.N2=1009) address as source address and type another detected NPD (Eg.N3=1005) as Dev1 and another NPD Dev2(Eg.N4=1003).
- Now type **NoDev** as '2'
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- This step will create another hop in network and we get the desired network configuration.
- Now reset the network and click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network which we configured .

By setting these parameters perform the experiment as mentioned in experiment No.2

Sr .no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	PacketTransmission
1.	50 ms	20	20	S	N	L
2.	100 ms	20	20	S	N	L
3.	150 ms	20	20	S	N	L

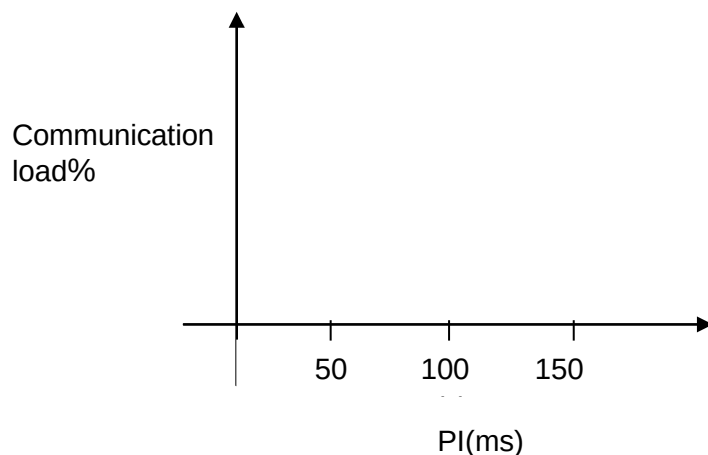
Sr .no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	PacketTransmission
1.	100 ms	20	20	S	N	L
2.	100 ms	20	20	S	A	L

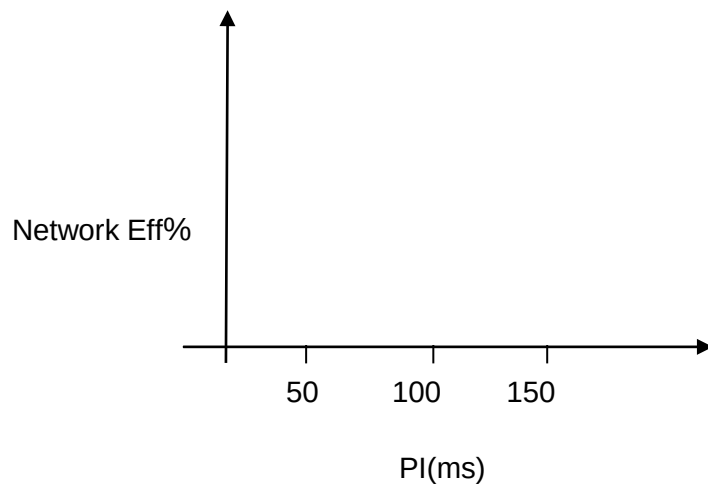
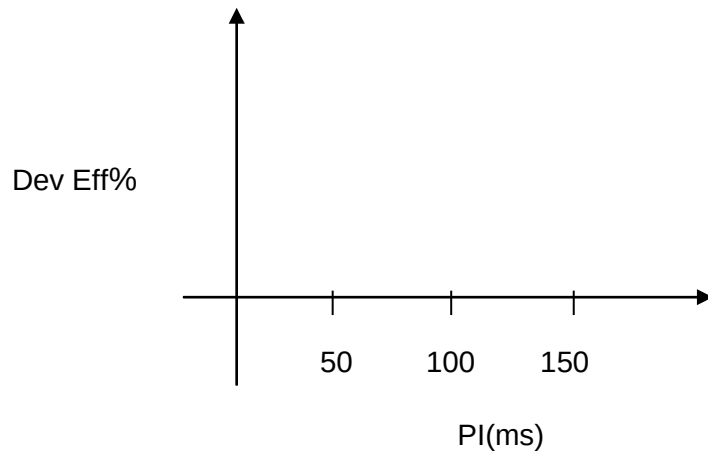
As,we have got the ResData on the (WSN Wireless parameter and Processing) Click on **Process** to get Communication Load , network efficiency, device efficiency

Observation Table for Tree Topology

PI	50ms	100ms	150ms
Communication Load%			
Device Efficiency%			
Network Efficiency%			

Now we will plot a Graph for Communication Load, Device Efficiency , Network Efficiency Vs PI



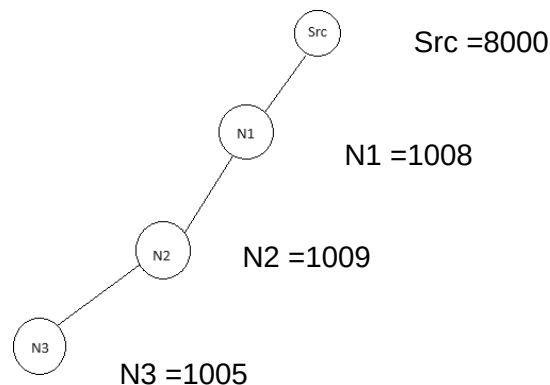


Observation Table for Tree Topology

Ack Type	N (No Acknowledgement)	A (Software Acknowledgement)
Communication Load%		
Device Efficiency%		
Network Efficiency%		

Case 3- Linear Topology

- Before we start Linear Topology Reset the network configuration



- Click on **NetworkConfig** and set as above.
- Now type Master address(8000) in source address, select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now let's create hops in the network,type source address as '8000' and type detected N1 device address in Eg.**Dev1=1008**.
- Now type **NoDev** as '1'.
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- Now reset the network and check hop is created or not.
- Now again click on **NetworkConfig**.
- Now select **SrcDev** (N1=1008) address as source address and type another detectedNPD (N2=1009) as **Dev1**.
- Now type **NoDev** as '1'
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.
- This step will create another hop in network.
- Now select **SrcDev** (Eg.N2=1009) address as source address and type another detected NPD (Eg.N3=1005) **Dev1** .
- Now type **NoDev** as '1'
- Again select apply 'Y' and Allow/Block 'A'. And then click on set filter control.

This step will create another hop in network and we get the desired network configuration.

- Now reset the network and click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network which we configured.

By setting these parameters perform the experiment as mentioned in experiment No.2

Sr .no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	50 ms	20	20	S	N	L
2.	100 ms	20	20	S	N	L
3.	150 ms	20	20	S	N	L

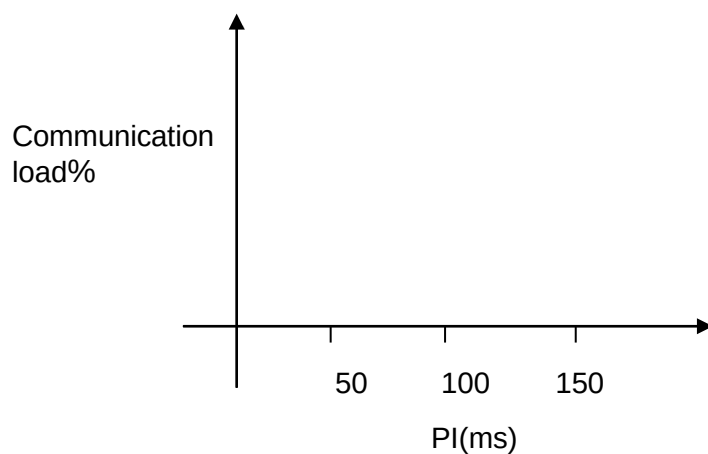
Sr .no	Packet Interval (ms)	Duration (sec)	Data Packet (Byte)	Packet Type	ACT Type	Packet Transmission
1.	100 ms	20	20	S	N	L
2.	100 ms	20	20	S	A	L

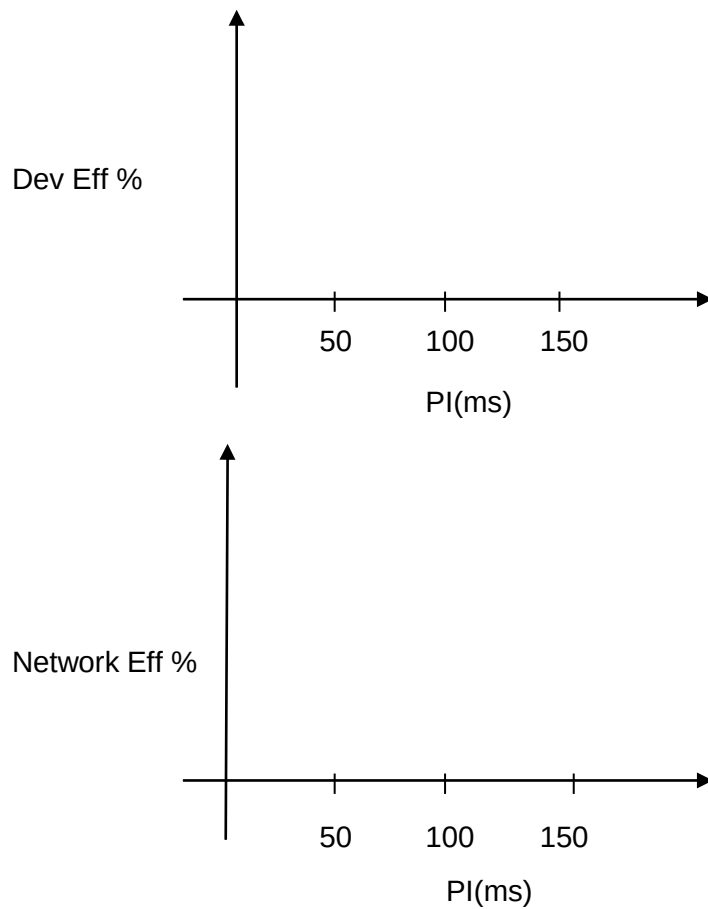
As,we have got the ResData on the (WSN Wireless parameter and Processing) Click on **Process** to get Communication Load , network efficiency, device efficiency

Observation Table for Linear Topology

PI	50ms	100ms	150ms
Communication Load%			
Device Efficiency%			
Network Efficiency%			

Now we will plot a Graph for Communication Load, Device Efficiency , Network Efficiency Vs PI



**Observation table for Linear Topology**

Ack Type	N (No Acknowledgement)	A (Software Acknowledgement)
Communication Load%		
Device Efficiency%		
Network Efficiency%		

Conclusion –

Here we verify different Communication Load , network efficiency, deviceefficiency for different topologies like Star, Semi tree, Tree.

EXPERIMENT NO. 10

EXPERIMENT NO: 10

NAME: -

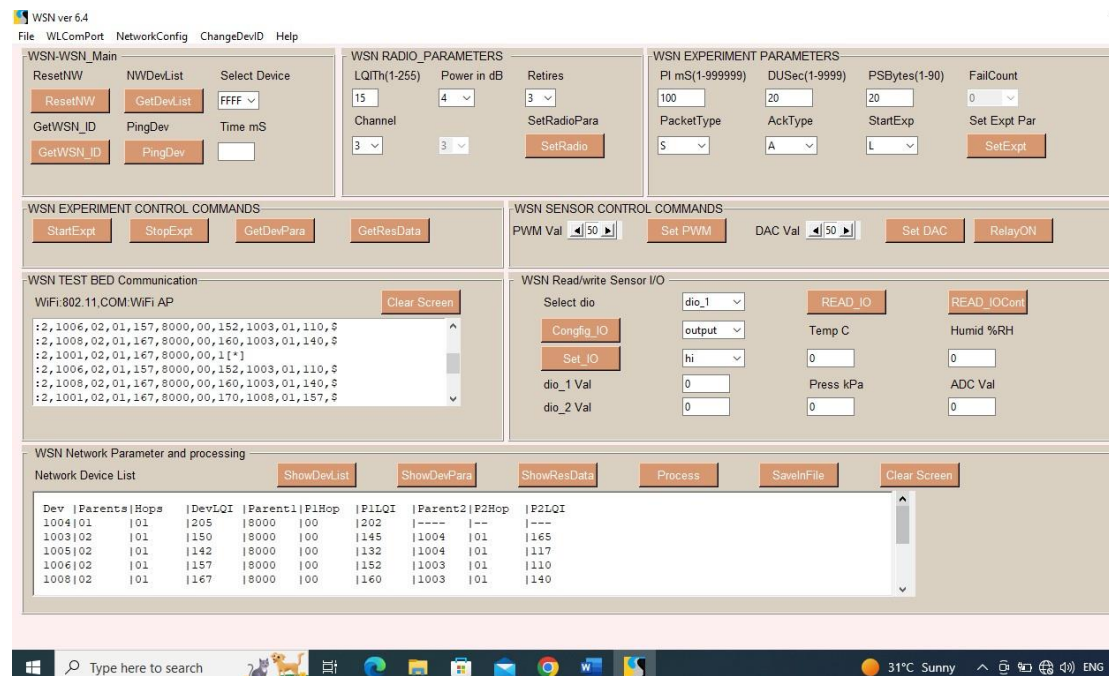
Application of Wireless Sensor Network to implement Network Nodes as IoT Sensors.

EQUIPMENTS: -

- WL-WSN NPD's 04 no's.
- Master / Gateway Trainer Kit 01 nos.
- PC or Laptop with WSN Application software.

PROCEDURE: -

- First switch on the Master / Gateway Trainer kit.
- Open GUI exe file WSN application exe.
- Now click on WLcomport and select SWIFI_GATEWAY.
- Click on the Connect tab.
- Click **OK**.
- Now We get the comport and Master device address on **WSN TEST BED Communication**.
- Now switch on all the nodes.
- Now click on **Get Device List** and **Show Device List**. So we will get the list of numbers of nodes connected in the network.



- For this experiment we are going to see how WSN is used in real life scenarios. Here we have Master Device working as a control room/base station and NPD as End devices. Here we can control PWM Signal, DAC Signal and Relay from master device to node and Also to get readings from all the Sensors (Temp/ Humidity/Pressure and ADC Value) on the nodes to the Control Room (MD).
- For this experiment we set the parameter from **WSN-WSN Main**.
- Click on **Select Device** to control particular Node (NPD) and select FFFF if you want to broadcast the specific command to all the Nodes (NPD).
- From the (WSN SENSOR CONTROL COMMAND) you can control the On and Off or brightness of the LED on the NPD by changing the PWMValue.
- Similarly, Now set DAC value on GUI (Click on **Set DAC (WSN SENSOR CONTROL COMMAND)**).
- Check the voltage on the NPD- DAC (CON2) using Digital Multimeter.
- Similarly you can Click on the RELAY ON Button , this will activate the RELAYS on the NPD.
- Click on **Read_IO (WSN Read/Write SENSOR I/O)** here we take inputs from the sensor nodes , to get the readings of temperature , humidity , pressure. To get continuous readings from the nodes Click on **READ_IOCont**.
- Now check Temp C, Humid %RH, Press kPa and ADC value displayed on the GUI.

For any External Device .

- There are 2 Channels – DIO 1 and DIO 2 which can be set as INPUT or OUTPUT.
- Once this is Set up Click on the **CONFIG IO**.

- If you have set the DIO as OUTPUT then You can set the DIO 1 and DIO 2 as Hi or Lo and then Click **SET IO**.
- Click on **Read_IO** (WSN Read/Write SENSOR I/O) and you will get the values dio_val1 and dio_val2 respectively which can be verified by connecting a DMM on Sensor Interface Conn Pin 1 and 2 respectively.

Conclusion-

In this experiment we understand how we can utilize network nodes as IoT sensor and how IoT devices are used for smart sensing and monitoring system.